Active Cartesian Modular Dynamics: Machine-Verified Foundations

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Repository: https://github.com/jeffgurd888/thet-logos

> "If we don't stop worrying about the Almighty dollar, we're not going to change the world!" — Caleb Gurd, as remembered by his father.

Version note. Version 1 (2026-10-15; DOI: 10.5281/zenodo.23408680) reported the program through Rung 8, together with the SM-Match program and the first round of the theories intake. Version 2 (2026-10-16; DOI: 10.5281/zenodo.23427964) added the closing of Rung 8's bosonic half (Rung 8B, §8B), fermionic second quantization (Rung 9, §9), bosonic second quantization (Rung 10, §10), and a fenced background-algebra appendix (Appendix A). Version 3 adds: the infinite-mode free theory (Pitch 2, §10A); interacting Hamiltonians at finite modes for both statistics, with the Pitch 3 verdict stated exactly as ruled (§10B); finite-dimensional Schrödinger dynamics (§10C); the discharge of Rung 7's boost window (§7 amendment); the graph-Laplacian bridge exhibit (Appendix A); and the opening stages of the continuum pitch (Pitch 4, §13A) — the lattice scoping, Stage 0, Stage 1a, and Stage 1b in its four sub-stages. It also carries a full re-audit of §16 (what is NOT claimed): two lines carried from Version 2 were overtaken by the work and are amended in the open (§0, corrections to Version 2). Everything else is carried from Version 2 unchanged except as amended in the open where dated amendment notes say so. Version 3 was published as a new version of the same record (DOI: 10.5281/zenodo.23444600). Version 4 (this version; prepared 2026-10-18) carries one open correction to Version 3 — the Pitch 3 verdict of §10B was superseded within a day of Version 3's publication and is corrected in §0, in the open, not silently — and adds §§18–28: the Yukawa interaction; the thermal clock; the XYZ screening wave's complete kill of the named sector grid; Pitch 4 Stage 2 and Stage 1c-0; the strong-continuity discharges (translations, one-particle Lorentz, Fock packaging, Fock-level Lorentz); the Schrödinger smalls and the Bridge completion; the decoherence toy, pre-compensation, and the resonance spectrum; the closed Round 2 intake (the OS re-pin, the Dixmier intake, the BRST intake); the unlock chain of co-import repairs; the Wightman re-pin; the trit experiment's pre-registered kill; and the vacuum-portrait exhibit. The program total at preparation is **3,276 theorem/lemma declarations** by the strict counting method of §0 (an attribute-inclusive count of the same modules reads 3,291). Version 4 is published as a new version of the same record; its DOI is minted on publication.

Abstract

This paper states exactly what the machine has verified in the Active Cartesian Modular Dynamics (ACMD) program through Pitch 4, Stage 1b, together with the SM-Match program and the first round of the theories intake. Every theorem named here is

proved in Lean 4 (tier T1), explicitly axiomatized with proved consequences (T2), stated as a labeled open target (T3/T5), or recorded as refuted. The program is machine-verified mathematical scaffolding toward a rigorous four-dimensional quantum field theory: a finite spectral triple and its classification; the Connes–Kreimer Hopf algebra of renormalization; the almost-commutative product with its twist derived rather than assumed; analytic and Hilbert-space infrastructure for a concrete model; verdicts on all six Wightman axioms in a one-mode fermionic model; genuine Schwartz test-function machinery and a geometric Poincaré action; a finite-dimensional Stone theorem and a joint functional calculus for commuting symmetric operators; a Fock-space inner product with proved creation/annihilation adjoints on both the fermionic and bosonic sides; fermionic second quantization — the functorial lift Γ(U), the differential lift dΓ, and a free Hamiltonian whose gap is derived at finite modes; the bosonic dΓ lift with its own derived free gap, and the bosonic Γ(U) lift, completed as a verified functor by the same-day campaign that proved the two permanent identities its functoriality had been priced at; a structural match against the Standard Model's gauge algebra, fermion content, classical Lagrangian sectors, and Higgs breaking pattern; and fit verdicts for KMS theory, classical Yang–Mills, Haag–Kastler algebraic QFT, and the finite spectral action. Version 3 adds: the free theory at countably many modes for both statistics, with the derived free gap in the eigenvalue-on-finite-particle-domain sense (Pitch 2); interacting Hamiltonians at finite modes — the fermionic campaign stands whole with gap persistence, and the bosonic campaign's sole priced clause, essential self-adjointness, was discharged the following day (Pitch 3 stands whole; §0 carries the correction to Version 3's conditional verdict, in the open); unitary Schrödinger dynamics with picture equivalence and proved time-invariance of the interacting gap clusters; full restricted-Lorentz invariance of the free one-particle mass-shell measure (the Rung 7 boost window, discharged); and the opening of the lattice road toward the continuum (Pitch 4): Wilson-action lattice gauge theory with exact gauge invariance, the Gibbs measure and correlation package, Elitzur's theorem at a general gauge group and every coupling, the reflection structure, the Osterwalder–Schrader form with positivity proved at zero coupling and reduced at general coupling to one named lemma, the OS Hilbert space, and the transfer matrix — with the lattice gap proved exactly at zero coupling and fenced, with its price, at general coupling. Version 4 adds: the first mixed fermion–boson (Yukawa-shape) interaction, with windowed gap persistence (§18); the thermal clock — every faithful finite state carries its modular clock, and for Gibbs states physical time is β × thermal time (§19); the complete screening of the XYZ triadic sector grid — all 24 named triples killed and the sector existence statement refuted, with Open Problem 2 itself untouched and open (§20); Pitch 4 Stage 2 (one exact blocking step; the qualitative constraint set proved closed under it; no quantitative uniform bound) and Stage 1c-0, PARTIAL (§21); strong continuity of the one-particle representation for both subgroups and its Fock-level packaging and Lorentz transport (§22); the Schrödinger exponential and Duhamel remainders and the Bridge exhibit's spectral completion (§23); the corpus's first partial trace, the pre-compensation reachability theorem, and the resonance-spectrum pin (§24); the closed Round 2 intake — the Osterwalder–Schrader re-pin, the Dixmier-trace intake (whose finite evaluation is identically zero, against the intake's own prediction), and the finite BRST intake (§25); the discharge of all three co-import fences, so the whole corpus imports into one environment (§26); the Wightman re-pin on genuine

tempered distributions, with the old existence statements — the 2026 pin's and Rung 4's sharpened bundle alike — proved vacuous and replaced by failable ones (§27); and two exhibits of a different kind: a pre-registered experiment that killed its own direction (the trit experiment, §28) and a numerical portrait of the Stage 0/1a vacuum (§28).

What this is not. No continuum limit is claimed. No interacting 4D quantum field theory is claimed. Interacting Hamiltonians are constructed (§10B) — at finite modes, for matter self-interaction only, with no gauge field; no interacting continuum theory of either statistics is constructed, and no scattering content appears anywhere. The Yang–Mills existence and mass gap problem is the program's stated horizon and mission target; it is not claimed, approached in the Clay sense, or implied by anything below. The mass gap appears in this paper only as a fixed, quarantined target statement against which future partial results can be measured; the gaps that are derived below (§§9, 10, 10A, and the lattice gap of §13A) are gaps of free Hamiltonians at finite or countably many modes, of an interacting finite-mode Hamiltonian under a proved persistence threshold, and of a finite-lattice transfer matrix at zero coupling — labeled as such wherever they occur. **Parts of the official Clay Yang–Mills statement closed by this corpus: 0.**

Tier key. T1: proved in Lean 4, zero `sorry`s. T2: axiomatized, with proved (T1) consequences. T3: theorem stated, proof open, labeled in-file. T5: opaque, quarantined, or priced window — theorem-shaped, content not established. KILL: refuted or withdrawn; returns only as a new entry by proof.

Verification status. All verification claims for Rungs 1–8 were re-checked on 2026-10-15 (§0). That re-check raised two flags — the finite spectral-action proofs failed to compile under the pinned toolchain (§12) and the Morley gauntlet file could not be rebuilt to completion in that pass (§13) — and **both flags were resolved on 2026-10-15**: `SpectralActionFinite.lean` was repaired (proof-local fixes only; no statement changed; no `sorry`/`admit`/axiom added) and its FIT verdict stands restored (§12), and `EdgeSignature.lean` was re-verified by a standalone green build (exit 0; zero `sorry`/`admit`; axiom checks on the 13 headline theorems clean; §13). The material added in Version 2 (§§8B, 9, 10, Appendix A) was verified at its landings on 2026-10-15/16 as reported in the §0 addendum. The material added in Version 3 (§§10A, 10B, 10C, 13A, the §7 amendment, and the Appendix A bridge exhibit) was verified at its landings on 2026-10-16/17 as reported in the §0 Version 3 addendum, where every count in this paper is re-derived from the current sources by a single stated method. The material added in Version 4 was verified at its landings on 2026-10-17/18 and is inventoried the same way in the §0 Version 4 addendum; every count in Version 4 was re-derived from the current sources by the same single method before this version was assembled.

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0. Verification protocol and corpus inventory

Counting method (uniform, reproducible). All counts in this paper are counts of `theorem` and `lemma` declarations, obtained by the regular expression `^\s*(private\s+|protected\s+)?(theorem|lemma)\s` applied to every file in `lean/ThetLogos/`. Counts include

auxiliary lemmas; they are counts of declarations, not of "importance." **Inventory freeze:** the corpus was inventoried at commit `790c1e0` on 2026-10-15. Work landing after the freeze (a Rung 9 start and a ternary substrate module, both in progress under separate directives) is outside this paper's scope and is not assessed here.

Corpus totals (exact).

Scope	Files	Theorem/lemma declarations
Whole Lean corpus (`lean/ThetLogos/`)	77	**2,016**
ACMD program files (Rungs 1–8, SM-Match, intake, Morley, targets/dossiers)	64	**1,352**
Earlier exploratory campaigns in the same repository (CF-kernel arc, exotic sectors, chromatic and related work)	13	664

The figure "300+" that circulated in earlier drafts of program summaries was never derived from a count of this corpus. It is withdrawn and replaced by the exact counts above.

Build (2026-10-15, reported exactly). `cd lean && lake build` completes successfully (3,324 jobs). Stated plainly: the root module `ThetLogos.lean` imports only 14 core modules, so the remaining 63 modules were additionally built one by one (`lake build ThetLogos.<Module>`). Result: **74 of 77 modules build green.** The three exceptions:

- `SpectralActionFinite.lean` **failed** at the 2026-10-15 re-check: four proof errors (`simp` made no progress / failed rewrite at lines 942, 1006, 1024, 1074), all inside the a_2 trace-computation proofs and consistent with Mathlib deprecation drift since the file's 2026-10-11 verification. No statement was changed and no `sorry` introduced; the proofs no longer closed under the pinned toolchain. **Repaired the same day (2026-10-15): proof-local fixes only; the file builds green again and the FIT verdict of §12 stands restored** (theories-intake dossier amended accordingly).
- `SpectralActionBridge.lean` failed transitively (it imports the file above); restored with the repair.
- `EdgeSignature.lean` (the Morley gauntlet file, §13): two rebuild attempts on 2026-10-15 timed out (10 and 25 minutes, under heavy machine contention) without emitting any error — the build did not finish in that pass. **Re-verified later on 2026-10-15 by a standalone build: exit 0, zero `sorry`/`admit`, axiom checks on the 13 headline theorems clean. Closed.**

A minor wart, recorded for completeness: two declaration names collide across modules in the shared `ThetLogos` namespace (`vacuum_cyclic` in `FreeField` and `FermionicFock`; `Dprod` in `ProductTriple` and `CausalProduct`), so those module pairs cannot be imported into a single environment. Each module builds and verifies on its own. Wiring the newer modules into the root import list, and repairing the spectral-action proofs, are queued mechanical follow-ups.

Sorry hygiene (exact). The corpus contains exactly **seven** `sorry`s, all pre-existing, all labeled in-file, none introduced by Rungs 1–8, SM-Match, or the intake:

Location	Count	Tier	Content
`BlockedQuestions.lean` (lines 76, 108, 140, 194, 274)	5	T5	Open conjectures Q1–Q5, filed as questions
`ModularTime.lean:206`	1	T3	`kms_identity_diagonal`; numerically verified, proof open
`XYZTripotents.lean:492`	1	T5	`xyz_tripotents_exist`; single documented target

There is no `admit` anywhere in the corpus. Every other occurrence of the word "sorry" in the sources is documentation prose, mostly statements that a file contains none.

Axiom footprint (spot-checked 2026-10-15). `#print axioms` was run on thirteen headline results. Eleven depend only on Lean's standard axioms `[propext, Classical.choice, Quot.sound]` — no `sorryAx`: `antipodeAxiomEval_proved`, `modularFlux_selfAdjoint`, `w5_span_true`, `wick_four_point_operator`, `schwartzComplete_discharge`, `jointFC_spectralMapping`, `higgs_stabilizer_is_Q`, `gibbs_kms_weight`, `stone_finiteDim`, `ann_adjoint`, `adag_adjoint`. The two labeled exceptions behave exactly as their labels say: `pauliBounded` depends additionally on the one T2 axiom `pauli_compactResolvent`, and `trace_collapse_hypothesis_fails` on the `CausalProduct` file-level T2 product axioms. And `kms_identity_diagonal` depends on `sorryAx` — precisely its T3 label.

v2 addendum (2026-10-16): increments since the Version 1 freeze

The Version 1 freeze above stands as the 2026-10-15 record. The work that landed after it — and that Version 2 reports — is inventoried here by the same counting method, as increments.

Program total (Version 2, decomposed). Version 2's program total, by the same decomposed method as the §0 table, is **1,648**

Increment	Files	Theorem/lemma declarations (definitions counted separately)
Rungs 1–8 era (Version 1 freeze)	64 program files	1,352
Rung 9 — fermionic second quantization (§9)	2	+50 (plus 13 definitions)
Rung 8B — bosonic assembly (§8B)	1	+31 (plus 6 definitions)
Rung 10 — bosonic second quantization (§10)	2	+31 (plus 13 definitions)
Background algebra — sexagesimal clock (Appendix A)	1	+31 (plus 12 definitions, 1 structure)

Increment	Files	Theorem/lemma declarations (definitions counted separately)
Γ functor campaign — Rung 10 Task 3 closed (§10)	1	+42 (plus 8 definitions)
Background algebra — flux-twist toy (Appendix A)	1	+38 (plus 10 definitions)
Background algebra — weld duality (Appendix A)	1	+73 (plus 18 definitions)
Program total (decomposed)		**1,648**

The ternary substrate module of Appendix A (`TritGrading.lean`, 350 lines) also landed after the Version 1 freeze; it is reported qualitatively in Appendix A and carries no separate increment in the decomposed chain above.

Builds (as reported at each landing). The root `lake build` was green (3,325 jobs, exit 0) at the Rung 9, Rung 8B, and Rung 10 landings, with the pre-existing corpus untouched. The newer rung modules are not in `ThetLogos.lean`'s root import list — the same staging as at Version 1 — and are verified by targeted builds, each green: `ThetLogos.SecondQuantGamma` and `ThetLogos.SecondQuantDGamma` (Rung 9, within the root build); `ThetLogos.BosonicAssembly` (2,452 jobs); `ThetLogos.BosonicSecondQuant` (2,453 jobs); `ThetLogos.BosonicGamma` (2,454 jobs); `ThetLogos.SexagesimalClock` (3,118 jobs, zero errors, zero warnings); `ThetLogos.FluxTwist` (3,120 jobs); `ThetLogos.WeldDuality` (3,120 jobs). Wiring the newer modules into the root import list remains a separate, queued decision.

Sorry hygiene (increments). The new files introduce no `sorry` and no `admit`: zero-sorry sweeps are reported clean for both Rung 9 files, for `BosonicAssembly.lean`, for both Rung 10 files, for `TritGrading.lean`, for `SexagesimalClock.lean` (which also has no `native_decide`), and for `FluxTwist.lean` and `WeldDuality.lean` (zero `sorry`/`admit`/`native_decide` in code for both). The corpus total of §0 — exactly seven, all pre-existing and labeled — is unchanged.

Axiom footprint (increments). `#print axioms` spot-checks on the new headline results return exactly `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, no new axioms: every headline theorem of both Rung 9 files (per that rung's dossier); eleven named headlines of Rung 8B, including `adjoint_relation`, `ccr_mixed`, `numberOp_basis`, `sector_orthogonal`, and `cycSub_eq_top`; fifteen named headlines of Rung 10 across its three tasks, including `dgammaBos_adjoint`, `freeHamiltonianBos_eigenvalue_gap`, and `gammaBos_apply_basis`; and eight named headlines of the sexagesimal clock module, including `crtEquiv` and `ClockGrading.reduceToTrit`; six named headlines of the flux-twist module, including `twistedShift_unitary` and `twistedShift_mul_gaugeMatrix`; and seven named headlines of the weld module, including `trace_weldPlus`, `etaCorr_sq`, and `signature_etaDirac`. The clock module's `lcm_one_through_six` depends on no axioms at all.

v3 addendum (2026-10-17): increments since the Version 2 freeze, counted by one method

The Version 2 record above stands as published. The work that landed after it — and that Version 3 reports — is inventoried here. For this addendum every count was **re-derived from the current sources** (at commit `4380a2a`) by a single method, stated exactly so any reader can reproduce it:

Counting method (Version 3 — strict line-initial, retained). A theorem or lemma is counted when a line of the comment-stripped source begins at column 0 with `theorem` or `lemma`: the regular expression `^(theorem|lemma)` applied after stripping Lean comments (line and nested block comments). This is the Version 2 §0 pattern, retained unchanged by the author's decision of 2026-10-17, taken after both candidate rules had been computed and compared (see the recount note below): the strict rule is the one whose total any reader can reproduce with the simplest possible grep, and it never overcounts. Definitions are counted by the analogous line-initial pattern over `def`/`abbrev`/`structure`/`instance`, admitting the standard same-line prefixes (an attribute, `noncomputable`, `private`, `protected`) that definition keywords routinely carry; the program total itself is the theorem/lemma count. The recount consequences follow, recorded here rather than absorbed silently:

- **The Version 2 total stands.** Recounting the Version 2 scope by the strict rule gives exactly its published **1,648**, and the Rung 9 increment is 50 as published in §0. Fifteen declarations in today's program modules carry a same-line `@[simp]` attribute prefix and are therefore **not** counted by the strict rule; an attribute-inclusive count of the same modules totals 2,432. The fifteen, all present and verified since their files landed: `gamma_apply` (Rung 9, §9); `loopEnd_nil`, `loopEnd_cons`, `loopProd_nil`, `loopProd_cons` (Pitch 4 Stage 0 — its increment is 68, as its landing dossier stated); `fermConfigEmb_apply`, `fermConfigEmbLim_apply`, `configEmb_apply`, `configEmbLim_apply` (Pitch 2 Campaign A — its increment is 57 line-initial, not the attribute-inclusive 61 of its dossier); and `boostLin_apply_zero`, `boostLin_apply_one`, `boostLin_apply_two`, `boostLin_apply_three`, `boostLin_symm_apply_zero`, `boostLin_symm_apply_one` (the Rung 7 boost discharge — 80 line-initial, not the attribute-inclusive 86 of its dossier).
- **The running chain is corrected.** The decomposed chain published in the repository's README at the time of assembly totaled 2,427, because it summed the landing dossiers' figures under their two historical patterns: the Pitch 2 and MassShellBoost dossiers counted attribute-inclusive (+4 and +6 over strict), the rest counted strict. Under the single strict rule the program total is **2,417**, and Version 3 reports that figure everywhere.

Program total (Version 3, decomposed).

Increment	Files	Theorem/lemma declarations (definitions counted separately)
Rungs 1–8 era (Version 1 freeze)	64 program files	1,352

Increment	Files	Theorem/lemma declarations (definitions counted separately)
Rung 9 — fermionic second quantization (§9)	2	+50 (plus 13 definitions)
Rung 8B — bosonic assembly (§8B)	1	+31 (plus 6 definitions)
Rung 10 — bosonic second quantization (§10)	2	+31 (plus 13 definitions)
Background algebra — sexagesimal clock (Appendix A)	1	+31 (plus 13 definitions and structures)
Γ functor campaign — Rung 10 Task 3 closed (§10)	1	+42 (plus 8 definitions)
Background algebra — flux-twist toy (Appendix A)	1	+38 (plus 10 definitions)
Background algebra — weld duality (Appendix A)	1	+73 (plus 18 definitions)
Subtotal — Version 2 scope (strict recount = as published)		1,648
Pitch 2 — free theory at countably many modes (§10A)	5	+137 (plus 57 definitions)
Pitch 3 Campaign 1 — fermionic interaction (§10B)	1	+58 (plus 15 definitions)
Schrödinger dynamics (§10C)	1	+49 (plus 10 definitions)
Pitch 3 Campaign 2 — bosonic interaction (§10B)	1	+69 (plus 21 definitions)
Rung 7 boost window discharged (§7 amendment)	1	+80 (plus 15 definitions)
Background algebra — bridge exhibit (Appendix A)	1	+29 (plus 7 definitions)
Pitch 4 Stage 0 — finite lattice gauge theory (§13A)	1	+68 (plus 33 definitions and instances)
Pitch 4 Stage 1a — correlations and Elitzur (§13A)	1	+68 (plus 13 definitions)
Pitch 4 Stage 1b-A — reflection structure (§13A)	1	+67 (plus 16 definitions)
Pitch 4 Stage 1b-B — OS form and positivity (§13A)	1	+44 (plus 10 definitions)
Pitch 4 Stage 1b-C — OS Hilbert space (§13A)	1	+42 (plus 17 definitions)
Pitch 4 Stage 1b-D — transfer matrix (§13A)	1	+58 (plus 13 definitions)
Program total (decomposed, strict line-initial method)		**2,417**

As at Version 2, the ternary substrate module (`TritGrading.lean`, 27 declarations) is reported qualitatively in Appendix A and carries no separate increment in this chain. For reference only, the whole Lean corpus as it stands today — 103 files, including the earlier exploratory campaigns and every post-freeze module whether or not it belongs to the program chain — counts **3,086** declarations by the strict rule; that figure is an inventory of the tree, not a program total, and is not a partition of the chain above.

Two raw-grep disclosures, so the counts can be checked without surprise: in `FermionicInteraction.lean` a raw line count gives 59 hits — the 59th is the word "theorem" opening a line inside the §4 header comment, disclosed in that campaign's dossier; the comment-stripped count is 58. In `SchrodingerDynamics.lean` the raw count gives 52 hits, of which exactly 3 are header or comment prose, disclosed in its dossier; the count is 49.

Builds (as reported at each landing, parent-verified). Every Version 3 module was verified by a targeted build (`lake build ThetLogos.<Module>`), each green, at its landing: Pitch 2 Campaign A modules (2,456 jobs) and Campaign B (3,207 jobs across the pitch modules); `ThetLogos.FermionicInteraction` (3,297 jobs); `ThetLogos.SchrodingerDynamics` (3,319 jobs); `ThetLogos.BosonicInteraction` (2,704 jobs targeted); `ThetLogos.MassShellBoost` (3,463 jobs); `ThetLogos.BridgeLaplacian` (3,117 jobs); `ThetLogos.LatticeGauge` (3,390 jobs); `ThetLogos.LatticeCorrelations` (3,391 jobs); `ThetLogos.LatticeReflection` (3,392 jobs); `ThetLogos.LatticePositivity` (3,393 jobs); `ThetLogos.LatticeOSSpace` (3,550 jobs); `ThetLogos.LatticeTransfer` (3,551 jobs). The newer modules are not in `ThetLogos.lean`'s root import list — the same staging as at Versions 1–2, disclosed in each module header; wiring them in remains a separate, queued decision.

Sorry hygiene (re-derived 2026-10-17). A comment-stripped sweep of all 103 corpus files finds the word `sorry` in code **exactly seven times** — the same seven labeled occurrences of the §0 table (five T5 in `BlockedQuestions.lean`, one T3 in `ModularTime.lean`, one T5 in `XYZTripotents.lean`) — and the word `admit` in code **zero times**. None of the Version 3 modules introduces either; every other occurrence of either word in the sources is prose, mostly statements that a file contains none. The corpus total of §0 — exactly seven, all pre-existing and labeled — is unchanged.

Axiom footprint (increments). `#print axioms` spot-checks on the Version 3 headline results return exactly `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, no new axioms: nine named headlines of Pitch 2 Campaign A and eight of Campaign B; `hamInt_gap_persistence`, `densityInt_not_dgamma`, and `intOp_isSymmetric` of Pitch 3 Campaign 1; ten named headlines of Schrödinger dynamics; six named headlines of Pitch 3 Campaign 2; six named headlines of the boost discharge (`boostInvariance_window_proved` among them); eight named headlines of the bridge exhibit; eight of Stage 0; the five scoped headlines of Stage 1a (`gibbsExpect_eq_integral`, `elitzur_single_link` among them); six of Stage 1b-B; eleven of Stage 1b-C; and eleven of Stage 1b-D (`transferOp_inner`, `transferOp_isSelfAdjoint`, `transferOp_zero_apply` among them).

v4 addendum (2026-10-18): increments since the Version 3 freeze, counted by the same method

The Version 3 record above stands as published. The work that landed after it — and that Version 4 reports — is inventoried here, by the same strict line-initial method of the Version 3 addendum (line-initial `^(theorem|lemma)` on comment-stripped source), every count re-derived from the current sources at assembly. The one-command recount any reader can run is in §17. The attribute-inclusive parallel figure for the same modules is **3,291** (the same fifteen `@[simp]`-prefixed declarations named in the Version 3 addendum account for the difference; no new attribute-prefixed declarations entered the chain).

Program total (Version 4, decomposed).

Increment	Files	Theorem/lemma declarations (definitions counted separately)
Subtotal — Version 3 scope (strict recount = as published)		2,417
Pitch 3's clause discharged — ESA-α and ESA-β (§0 correction 1; §10B)	2	+33 (plus 7 definitions)
Yukawa interaction — first mixed fermion–boson term (§18)	1	+119 (plus 44 definitions)
Thermal clock (§19; including the housekeeping §7 scalar-shift closure, +4)	1	+33 (plus 12 definitions)
Triadic pin (§20; definitions only — no theorems claimed)	1	+0 (4 definitions)
XYZ screening wave — the sector grid (§20)	1	+19 (plus 1 definition)
Pitch 4 Stage 2 — one exact blocking step (§21)	1	+60 (plus 15 definitions, 1 structure)
Translations strong continuity (§22)	1	+10
Pitch 4 Stage 1c-0 — β-analyticity exhibit, PARTIAL (§21)	1	+87 (plus 14 definitions)
Fock packaging — the Γ port to the assembled Fock spaces (§22)	3	+215 (plus 46 definitions)
Lorentz continuity, one-particle (§22)	1	+28 (plus 8 definitions)
Schrödinger smalls — exponential identification, Duhamel (§23)	1	+24 (plus 5 definitions)
Bridge completion — attainment, decay, graph-Faraday (§23)	1	+60 (plus 18 definitions)
Decoherence toy — partial trace and pinching (§24)	1	+37 (plus 6 definitions)
OS re-pin — Round 2 (§25)	1	+15 (plus 12 definitions)
Dixmier intake — Round 2 (§25)	1	+24 (plus 6 definitions)

Increment	Files	Theorem/lemma declarations (definitions counted separately)
BRST intake — Round 2 (§25)	1	+24 (plus 13 definitions, 1 structure)
Fock-level Lorentz continuity, both statistics (§22)	2	+14
Wightman re-pin (§27)	1	+14 (plus 11 definitions)
Pre-compensation (§24)	1	+28 (plus 4 definitions)
Resonance spectrum (§24)	1	+15 (plus 2 definitions)
Program total (decomposed, strict line-initial method)		**3,276**

Three repair landings in this period moved no count, as repairs must not: the `numberOp` rename, the six-name packaging rename (the unlock chain, §26), and the housekeeping bundle's Part A (seven stale module headers corrected in place, comment-only; its Part B is the ThermalClock +4 in the row above). For reference only, the whole Lean corpus as it stands today — 128 files, including the earlier exploratory campaigns and every module whether or not it belongs to the program chain — counts **3,945** declarations by the strict rule (3,999 attribute-inclusive); that figure is an inventory of the tree, not a program total.

Builds (as reported at each landing, parent-verified). Every Version 4 module was verified by a targeted build (`lake build ThetLogos.<Module>`), each green, at its landing: `BosonicESA` and `BosonicESABeta` (2,706 jobs at β); `YukawaInteraction` (3,313); `ThermalClock` (3,310); `TriadicScreening` (3,129); `LatticeBlocking` (3,558); `TranslationContinuity` (3,393); `LatticeAnalyticity` (3,530); the Fock packaging modules (3,675 across the brick); `LorentzContinuity` (3,661); `SchrodingerSmalls` (3,497); `BridgeCompletion` (3,316); `DecoherenceToy` (3,244); `OSRepin` (3,783); `DixmierTrace` (3,228); `BRSTFinite` (3,117); `FockLorentzBos` (3,672) and `FockLorentzFerm` (3,669); `WightmanRepin` (3,810); `PreCompensation` (3,245); `ResonanceSpectrum` (3,317); and `FockCoImport` (3,681), the single-environment certificate of §26. The newer modules are not in `ThetLogos.lean`'s root import list — the same staging as at Versions 1–3, disclosed in each module header.

Sorry hygiene (re-derived 2026-10-18). A comment-stripped sweep of all 128 corpus files finds the word `sorry` in code **exactly seven times** — the same seven labeled occurrences of the §0 table (five T5 in `BlockedQuestions.lean`, one T3 in `ModularTime.lean`, one T5 in `XYZTripotents.lean`) — and the word `admit` in code **zero times**. None of the Version 4 modules introduces either. The corpus total of §0 — exactly seven, all pre-existing and labeled — is unchanged.

Axiom footprint (increments). `#print axioms` spot-checks on the Version 4 headline results return exactly `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, no new axioms — across the landings: the ESA headline `bosonicESA_proved`; five Yukawa headlines; nine ThermalClock headlines and its four §7 additions; all 19 screening theorems; 13 Stage 2 headlines; the translations cap; nine Stage 1c-0 headlines; the packaging headlines in both statistics; the Lorentz-continuity headlines at one-particle and Fock level; seven smalls headlines; nine Bridge-completion headlines; 13

decoherence-toy headlines; 12 OS re-pin headlines (its three correction theorems depend on no axioms at all); ten Dixmier headlines; ten BRST headlines; 13 Wightman re-pin headlines (its existence-vacuity correction likewise axiom-free); nine pre-compensation headlines; and all 15 resonance-spectrum headlines.

Corrections to Version 2 (in the open)

Version 2 is a dated snapshot and is not retro-edited. Four of its statements are overtaken by the work Version 3 reports, and the recount above corrects the running chain built since Version 2. They are listed here, and amended in place in the sections named:

1. **§16, "No infinite-mode or continuum Γ" — FALSE since 2026-10-16.** Pitch 2 (§10A) verifies Γ at countably many modes for the free theory. The line is amended in §16; the continuum clause of it stands.
2. **§16 and the Version 2 abstract, "no interacting theory of either statistics is constructed" — overtaken by Pitch 3 (§10B).** Interacting Hamiltonians at finite modes are constructed for both statistics. What stands: no interacting continuum theory, no gauge-coupled interaction, no scattering. The lines are amended in the abstract and §16.
3. **§7's mass-shell track and §14 item 3 — the priced boost lemma is discharged.** The one-dimensional Jacobian identity is proved and the window closed whole (§7 amendment); §14 item 3 is marked discharged.
4. **§14 item 2 — the infinite-mode limit is discharged in the free theory** by Pitch 2 (§10A); the item is amended to its priced remainder (the spectral gap statement; rectangular functoriality at Task 3b scope).
5. **The counts stand as published.** Under the strict line-initial rule retained for Version 3, the Version 2 program total recounts to exactly its published 1,648, and the Rung 9 count is 50 as printed. What the recount corrects is the running chain assembled since (see the addendum): two landing dossiers counted attribute-prefixed declarations; the strict full-program total is 2,417.

Corrections to Version 3 (in the open)

Version 3 is a dated snapshot and is not retro-edited. Three of its statements are overtaken by the work Version 4 reports. They are listed here, and amended in place in the sections named:

1. **§10B's Pitch 3 verdict — the bosonic campaign's sole priced clause is DISCHARGED.** Version 3 reported Pitch 3 as standing for fermionic interaction whole and for bosonic interaction except essential self-adjointness, the closure equality `hamPMap.closure = hamPMap†` fenced on the analytic input the pinned Mathlib lacks (in Version 3's route analysis, on von Neumann's double-adjoint theorem `T†† = T.closure`, which is absent from the pinned Mathlib and remains absent). Within a day of Version 3's publication, ESA-α reduced the clause to the single inclusion `(hamPMap)† ≤ (hamPMap).closure` (its equivalence and implication directions proved; the remaining direction fenced), and ESA-β **proved that**

inclusion directly — by exact sector-preserving truncation: for an adjoint graph pair, the occupation-cutoff truncations are exact graph pairs at every cutoff, so the pair lies in the closure of the graph — with **no parameter hypotheses**: `BosonicESA` holds for every mode count, frequency tuple, quartic tensor, and real coupling. The von Neumann identity is not used and stays fenced. **Pitch 3 stands whole at finite modes, for both statistics, with self-adjointness proved.** The fences that do stand are unchanged: gap persistence for the bosonic interaction is windowed/sectorwise, and uniform bosonic persistence remains disproved in-module (§10B, §15). §16's scoped-senses line is amended accordingly. The exact diagonal gap's surviving fence is likewise unchanged (strong dichotomy only; the weak form is false).

2. **§14 items 9, 10, 14, 15, 16 — discharged or narrowed.** Item 9 is correction 1 above. Item 10's `adag` repair and the Yukawa campaign on top of it both landed (§18, §26) — and the repair exposed two further co-import fences the Version 3 text did not know (`numberOp`, then the packaging six), all three now discharged (§26). Item 14's Schrödinger smalls landed whole (§23). Item 15's Bridge follow-ons landed whole (§23). Item 16's decoherence toy landed (§24) — its premise ("no partial trace exists in the corpus or the pinned Mathlib") was re-verified at the build and the partial trace was built from scratch.
3. **§12's Round 2 note — Round 2 is now CLOSED.** The note recorded the OS fit-audit half as killed and the re-pinning half as standing priced. The re-pin landed (§25), and with it the Dixmier and BRST intakes (§25). Two of the re-pin's findings correct statements Version 3 carried without flagging: the 2026 OS and Wightman existence statements are **proved vacuous** (inhabited by an all-`True` instantiation), and Rung 4's sharpened `SharpWightmanTarget` bundle is vacuous as an existence form for the same reason one level up (§27). The six sharp definitions as conditions, and every Rung 5 verdict against them, are **not** impugned; what is superseded is the `∃`-bundling.

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1. Rung 1 — The finite spectral triple (T1) — STANDS

The finite factor is a concrete matrix spectral triple; its classification and constraint census are machine-checked.

- **Classification and census.** The generator family `smGen : Fin 12` gives 144 pairs; the order-zero condition is checked machine-side over 147,456 complex components (`order_zero_comm`). Three-generation inheritance is proved in `ThreeGen.lean` (20 declarations), including `smGen3_order_zero` and the Kronecker-factorization identities. The 576-pair order-one census is a T4 Python scan, not a Lean proof.
- **Modular dynamics.** `thermal_flux_vanishes`: the modular flux vanishes identically on Gibbs states for every inverse temperature β. `exists_activeDriver`: a non-equilibrium driver producing nonzero flux exists. `modularFlux_selfAdjoint`: the flux φ = i[K, D] is Hermitian. Gauge covariance and trace gauge-invariance of the flux are proved.

- **Ultimate reduction.** `UltimateReduction.lean` proves 6 T1 theorems, including `dynamicallyFrozen_of_noHeat_noGauge` and `equilibriumKillsFlux`. The same file records finite-level kills: micro-vortices (no vorticity operator exists in the finite triple) and the vortices→matter→gravity chain (needs continuum data).
- **Connes–Kreimer Hopf algebra.** `Renormalization.lean` (60 declarations) formalizes the Hopf algebra of rooted trees: coproduct, B_+ grafting, counity, coassociativity. The record on the antipode is two-sided and both sides are kept: the as-stated antipode axiom was **refuted** (`antipodeAxiom_false`), and the corrected evaluation version was **proved** (`antipodeAxiomEval_proved`, T5→T1). The refuted statement is not laundered into the proved one.

2. Rung 2 — The almost-commutative product (T1) — STANDS

`ProductTriple.lean` (63 declarations) builds the product of an axiomatized manifold factor with the finite triple.

- **Product Dirac identity.** `Dprod_sq`: $D_{prod}^2 = D_M^2 \otimes 1 + 1 \otimes D_F^2$. The formerly axiomatized `product_sum_of_squares` was promoted to a proved T1 — upgraded, not resurrected.
- **Twist derived, not assumed.** The product twist is the flip on the doubled product algebra (`flipProd_involutive`, `flipProd_CM_regular`); the candidate $\rho_M \otimes \rho_F$ product twist was killed (it does not typecheck).
- **Fluctuation decomposition.** `prod_commutator_decomp` splits inner fluctuations into manifold and finite parts.
- **KO data.** `koProd_signs`: KO signs $\langle -1, 1, -1 \rangle$ (`koDim_add`: $(4 + 6) \% 8 = 2$).
- **Causality re-entry.** `CausalProduct.lean`: the finite-level Span-5 causality obstruction does not carry to the product (`trace_collapse_hypothesis_fails`, `product_not_forced_central` — T1 consequences of the T2 product axioms, footprint labeled in §0). Genuine causal order is pinned at its exact price: global hyperbolicity.
- **The 6-dimensional caveat.** The surviving finite represented gauge algebra is six real dimensions: Option-A kills the represented $M_3(\mathbb{C})$ summand. The full Standard Model gauge group is not claimed from the finite factor. This caveat stands and is revisited precisely in §11.

3. Rung 3 — Lorentzian-structure pins (T2+T1, one partial) — STANDS

- **Antilinear J — promoted (T2+T1).** The obstruction (bare $\mathbb{C}$-modules; J not even statable) was removed by explicit enrichment of `TripleData` (sesquilinear form; positivity, nondegeneracy, completeness explicitly not assumed; antilinear `Janti` with KO axioms). Product theorems proved T1: `Jprod_sq` ($J^2 = -1$), `Jprod_D` ($JD = DJ$), `Jprod_Γ`.
- **Krein inner product — promoted (T2+T1).** Fundamental symmetry η with $\eta^2 = 1$ and form-self-adjointness proved (`ηprod_sq`, `ηprod_selfadj`). No adjoint

operator is constructed and no spacetime signature is claimed: η is an abstract fundamental symmetry.

- **Twisted first-order condition — PARTIAL.** The opposite representation is built in the `conjJ`-shadow of a° = Ja*J⁻¹ (multiplication reverses, T1); the diagonal reduction is proved (`prodTwistedFOC_diag_reduce`). The exact remaining obstruction is priced: no `Star` instance on `Module.End` over the tensor module in this setting, and factor-level order-one is absent from the abstract `TripleData`.
- **Continuum scoping — priced roadmap, no limit claimed.** "Finite triple converges" language is killed with reason; the continuum is scoped per-axiom as future work.
- **Promotion honesty.** The Rung 3 promotions are T2-axiomatization plus T1 consequences — not derivation from analysis. This paper repeats that qualification wherever the promotions are cited.

4. Rung 4 — The analytic house (T1 + one T2 bridge) — STANDS

Directive: "Build the house and use the gaps as windows."

- **Bounded-triple instance for the concrete Pauli model.** `pauliBounded : BoundedAnalyticTriple` forms. Its mathematical content is proved T1 underneath (`pauliD_sub_det`, `pauliD_resolvent`, `pauliD_resolvent_compact` — finite-dimensionality is the whole proof of compactness); one model-specific T2 axiom (`pauli_compactResolvent`) bridges the deliberately opaque `HasCompactResolvent` predicate. The window is eliminable only by redefining that predicate transparently — a design decision, priced in the dossier. The model stays analytically trivial (scalar representation, vanishing commutators); it discharges analytic debt, not gauge content.
- **Hilbert completion and bounded *-representation (T1).** Completion of the model is trivial (`CompleteSpace (EuclideanSpace ℂ (Fin 2))`); the bounded *-representation of the Pauli algebra is proved with the true adjoint star, transported from matrix conjugate-transpose (`repScal_is_star_rep`). The incorrect `star := conjJ` construction was explicitly refused earlier in the program and stays refused.
- **Smeared field operators (14 T1).** Creator-half canonical (anti)commutation relations for smeared fermionic and bosonic fields, smeared Pauli exclusion, and one-point vanishing (`FieldOperators.lean`).
- **Wightman targets framed (T5, unproved by design).** All six axioms are stated as structured propositions with exact hypotheses, conclusions, and per-axiom prices (`WightmanTargets.lean`). Stating them is not proving them; Rung 5 takes the verdicts.

5. Rung 5 — The Wightman ascent on a one-mode model — STANDS

On a one-mode fermionic Fock model ($\wedge(\mathbb{C}) \cong \mathbb{C}^2$, Majorana field $\psi(c) = c \bullet (a^\dagger + a)$), all six targets received T1 verdicts. These are verdicts about the toy model and about the framings — not about 4D QFT.

- **W1 (fields exist): PROVED** (`w1_majorana`).
- **W2 (covariance): PROVED for $\mathbb{Z}_2$ fermion parity** (`w2_parity`). This is explicitly not Poincaré covariance; the group is `ZMod 2` acting by parity.
- **W3 (spectrum condition): PROVED for the trivial translation representation** (`w3_trivial`) — a frame-consistency check (spectrum $\{0\} \subseteq$ forward light cone), not Stone's theorem.
- **W4 (locality): PROVED** at $s = +1$ (`w4_majorana`); the one-mode model has no spacetime, so the spacelike hypothesis is vacuous.
- **W5 (vacuum cyclicity): REFUTED as framed; corrected version PROVED.** Monomial-range density is false even for a free fermion (`w5_framed_false`); the linear-span density the literature means is true (`w5_span_true`). The Rung 4 framing was strictly stronger than the literature; the refutation is a framing correction recorded as a T1 result.
- **W6 (non-triviality): bosonic form proved but fermion-blind; corrected version PROVED.** The bosonic-form criterion gives $-2 \neq 0$ for the free Majorana field — a documented false positive (`w6_majorana`); the corrected Koszul-sign Wick criterion vanishes as it should (`w6_fermionic_wick`).

Rung 5 files: `OneModeFermion.lean` (37 declarations) and `Rung5W1–W6` (39 declarations).

6. Rung 6 — Infinite modes — STANDS

- **Schwartz machinery — premise correction, then GO.** The Rung 4/5 framing "Mathlib has no Schwartz space" was wrong: Mathlib ships the full Schwartz machine (Doll 2026). `SchwartzSpace.lean` proves 14 T1 results on genuine $\mathscr{S}(\mathbb{R}^4, \mathbb{C})$ — closure under derivatives and polynomial multiplication, temperate growth, Fourier inversion — and discharges the earlier T2 test-function marker. One genuine gap remained at this rung: no `CompleteSpace` $\mathscr{S}$`(E,F)` instance in Mathlib (discharged at Rung 7, §7).
- **Poincaré — geometric half GO.** `PoincareAction.lean` (11 declarations): the Lorentz group O(1,3) and Poincaré group as `Group` instances; the field action `f ↦ f(Λ⁻¹(x−a))` satisfies the group-action laws and is $\mathbb{C}$-linear in the test function (`poincareAct_mul`, `poincareActLin`). The one-particle unitary representation and second quantization stayed priced at this rung.
- **Wick combinatorics — PARTIAL.** `WickSigns.lean` (31 declarations): complete pairings of `Fin (2n)`, the Koszul sign, its independence of pair ordering, and the four-point signs (+1, −1, +1) (`koszulSign_four_point`). The operator Wick identity stayed priced (discharged at Rung 7 in a two-mode model).

- **Stone — analytic core T1.** `continuous_char_lift`: a continuous norm-one character of $(\mathbb{R},+)$ is `exp(i·a·t)` — the analytic core of Stone's theorem. The full finite-dimensional Stone theorem hit Lean engineering friction at this rung (discharged at Rung 7).

7. Rung 7 — The frontier — STANDS

Four of the five outstanding prices were discharged T1; the fifth was narrowed. This rung also marks where formalized analysis currently stops.

- **Schwartz completeness — T1.** `schwartzComplete_discharge` (`SchwartzComplete.lean`, 415 lines): $\mathscr{A}(\mathbb{R}^n, \mathbb{C})$ is Fréchet-complete — the genuine Mathlib gap from Rung 6, closed by a direct proof (countable uniformity basis → uniform convergence of all iterated derivatives → decay bounds pass to the limit).
- **Operator Wick identity — T1, with a correction.** In a concrete two-mode fermionic Fock model, `wick_four_point_operator`: $\omega(\psi_0\psi_1\psi_2\psi_3) = G_{01}G_{23} - G_{02}G_{13} + G_{03}G_{12}$, with all Wick hypotheses proved (CAR, Hilbert adjoints, quasi-free move), and the Koszul signs discharged against Rung 6. Correction recorded in-file: Rung 6's priced hypothesis pair $(\{\psi_i,\psi_j\} = 2G_{ij}$ together with $\omega(\psi_i\psi_j) = G_{ij})$ is jointly false in the model; the identity holds with the full non-symmetric G. The n-point expansion remains priced (T5).
- **Stone / joint spectral ONB — T1.** `jointSpectral_orthonormalBasis` (commuting symmetric family ⇒ orthonormal basis of joint eigenvectors) and the full finite-dimensional Stone theorem `stone_finiteDim` (strongly continuous unitary one-parameter group ⇒ skew-adjoint generator diagonal in that basis, `(U t)(b k) = exp(i·a k·t) • b k`).
- **Joint functional calculus — T1.** For commuting bounded symmetric operators, `SpectralFrontier.lean` proves a joint functional calculus stronger than Mathlib's single-operator continuous functional calculus, with the spectral mapping theorem `jointFC_spectralMapping`.
- **Mass-shell track — PARTIAL.** The invariant measure $d^3p/2E_p$ is constructed as a density-weighted pushforward; its integral formula and rotation invariance are proved, and the restricted (proper orthochronous) Lorentz group is formed as a subgroup. Boost invariance is priced at exactly one missing lemma: the one-dimensional Jacobian identity $|\varphi'| = E'/E$ for the x^1-boost. The one-particle representation law and unitarity-from-invariance are proved at function level; strong continuity and the linear-isometry packaging remain priced. Second quantization was blocked at this rung by the Fock inner-product gap — the subject of Rung 8.
- **Amendment (2026-10-17) — the boost window is DISCHARGED WHOLE.** The one missing lemma priced above is proved, and the cap the pricing flagged as likely is proved with it. `MassShellBoost.lean` (1,629 lines; 80 declarations, 15 definitions; targeted build green, 3,463 jobs; six headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The x^1-boost is built as a concrete element of the restricted Lorentz group (`boostRestricted`: Minkowski-form preserving, determinant 1, orthochronous); the shell parametrization is

intertwined with the boosted-energy identity $E' = \gamma(E - \beta u_1)$; the priced Jacobian identity is proved exactly as scoped — $|\varphi'(u_1)| = E_{\varphi(u_1)}/E_u$, with φ strictly monotone and surjective (`boostSlice_jacobian`, `boostSlice_abs_deriv`) — and the density cancellation $(E'/E)\cdot(2E')^{-1} = (2E)^{-1}$ assembles into `boost_preserves_massShellMeasure`: the boost's shell action is measure-preserving for the mass-shell measure. The cap: **every** restricted Lorentz transformation factors as (spatial lift)$^{-1}$ ∘ (x^1-boost) ∘ (spatial lift) — the spatial factor built from a Householder reflection, with the block lemma that a Lorentz map fixing the time direction is a spatial lift — so `boostInvariance_window_proved` holds for every Λ: **the free one-particle mass-shell measure is invariant under the full restricted Lorentz group.** Scope, exactly: the free, one-particle measure. This rung's other priced items are untouched by the discharge: **strong continuity** of the one-particle representation remains priced (repriced by the 2026-10-17 parked-fleet gauntlet: the translation part is a discharge-ready S — the file itself names the dominated-convergence lemma and the dominator $4\|\psi\|^2$; the Lorentz part is S–M, with this discharge's Jacobian identity supplying the domination), and **Fock packaging** remains priced at M (a concrete orthonormal-basis port; the abstract SymmetricAlgebra route stays killed, §15).

- **Amendment (2026-10-18) — this rung's priced remainders are DISCHARGED.** Strong continuity of the translation subgroup is proved (`TranslationContinuity.lean`, §22); strong continuity of the Lorentz subgroup at the one-particle level is proved for the full restricted Lorentz group (`LorentzContinuity.lean`, §22); Fock packaging is built by the concrete orthonormal-basis route (§22); and Fock-level Lorentz strong continuity is proved for both statistics (`FockLorentzBos`/`FockLorentzFerm`, §22). What remains open on this rung is exactly one statement: strong continuity of the joint action in a topology on the full Poincaré group — every continuity result in this corpus is subgroup-wise, and no joint topology on `RPoincare` is introduced anywhere (§14, as amended).

8. Rung 8 — The Fock inner product — STANDS (fermionic) / PARTIAL (bosonic, closed at §8B)

- **Fermionic — STANDS.** `FermionicFock.lean` (43 theorems, 1,064 lines): the Fock space as ℓ^2 over occupation subsets, with Pauli exclusion automatic from the subset guards. Proved T1: creation/annihilation adjoints (`ann_adjoint`, `adag_adjoint` — the fermionic discharge of the W0 inner-product gap), the canonical anticommutation relations, sector orthogonality, the number operator with eigenvalues `S.card`, and vacuum cyclicity. **Fermionic second quantization is unblocked.** Priced windows, stated in-file: unitary equivalence with exterior-power subspaces, the infinite-mode limit, and the general-n Gram determinant (base case and reassembly proved; inductive step priced).
- **Bosonic — PARTIAL at this rung; closed at §8B.** `BosonicFock.lean` (25 theorems): ℓ^2 over finitely supported occupation configurations, with creation/annihilation as linear maps on a dense domain (the honest standard formulation for unbounded operators). The inner product, orthonormal basis, explicit basis

actions, and the basis-level adjoint relation (`adjoint_basis` — the computational core of W0) are proved. The final linearity assembly of the full adjoint relation stalled on Mathlib API friction (submodule inner-product coercion handling) and is priced as engineering, not mathematics; the induction principle needed for the assembly is proved. CCR, number operator, and sector results follow from the basis actions by standard computations not yet formalized. All of these were discharged in full at Rung 8B (§8B), with no weakened statements.

8B. Rung 8B — Bosonic assembly (T1) — STANDS; the Rung 8 bosonic partial is closed

`BosonicAssembly.lean` (520 lines; 31 theorems and 6 definitions, zero sorrys; imports `BosonicFock.lean`, which is unmodified) performs exactly the assembly Rung 8 priced. All results are T1, on the dense finite-particle domain `Dom d` over a finite mode set `Fin d` — the free bosonic theory, stated with the same honesty labels as §8.

- **Full adjoint relation.** `adjoint_relation`: for all `x y : Dom d`, `⟪ann i x, y⟫ = ⟪x, adag i y⟫`, assembled from Rung 8's `adjoint_basis` by its own two-variable induction principle; the Rung 8 API friction yielded to an explicit motive. Companion form `adjoint_relation_adag`.
- **CCR on `Dom d`.** `ccr_mixed` — the mixed commutator `(ann i) ∘ₗ (adag j) − (adag j) ∘ₗ (ann i)` equals `1` if `i = j` and `0` otherwise, as linear maps on `Dom d` — together with `ccr_ann_ann`, `ccr_adag_adag` (annihilators commute; creators commute) and the pointwise form `ccr_mixed_apply`. Proved first on coefficient space and transported through the representation, not assembled by fiat.
- **Number operator.** `numberOp = ∑ i, (adag i) ∘ₗ (ann i)`, with `numberOp_basis`: the basis eigenvalue is the total occupation count, and `numberOp_vacuum`: the vacuum eigenvalue is `0`.
- **Vacuum and sectors.** `ann_vacuum` (annihilation kills the vacuum in `Dom d`); `sector_orthogonal`: sectors of distinct total occupation are orthogonal.
- **Vacuum cyclicity (stretch target, landed).** `cycSub_eq_top`: the submodule of `Dom d` generated from the vacuum by finitely many creation operators is everything.

Priced remainder, unchanged and not attempted here: bosonic Γ(U) (the subject of §10, Task 3); the infinite-mode limit; unitary equivalence with symmetric tensor-power sectors and the Gram/permanent formula (blocked upstream on n-fold tensor inner products); the continuum one-particle space and the mass-shell representation (the §9 Task 4 frontier).

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9. Rung 9 — Fermionic second quantization (T1) — STANDS (Tasks 1–3) / PARTIAL (Task 4)

All results in this rung are T1: 50 theorems and 13 definitions across two files, zero sorrys, with axiom spot-checks returning exactly `[propext, Classical.choice, Quot.sound]` for every headline theorem. Labels that apply to everything below: **free**

theory, finite modes. This rung is not a continuum QFT, not an interacting theory, and not the Yang–Mills mass gap; the one gap in it is derived, and it is labeled exactly where it lives.

- **Task 1 — the Γ lift: STANDS.** `SecondQuantGamma.lean` (992 lines, 32 theorems, 10 definitions). For a one-particle matrix `M` on the mode space `Fin d`, `Γ(M)` is constructed on the Rung 8 fermionic Fock space by the compound-matrix (minor) formula. Proved: `gamma_one` (`Γ(1) = 1`); `gamma_mul` (`Γ(M * N) = Γ(M) * Γ(N)`); `gamma_adjoint` (`Γ(M*) = Γ(M)*`, in inner-product form); `gamma_vacuum` (`Γ(M)` fixes the vacuum, for every `M`); `gamma_sector` (`Γ(M)` preserves every n-particle sector); `gamma_inner` with `gamma_comp_self` and `gamma_self_comp` — for unitary `M`, `Γ(M)` is unitary as an operator identity; and `gamma_diagonal_basis` — `Γ(diag v)` multiplies a basis vector by the product of the selected phases. The supporting determinant core is proved in the same file, from scratch where Mathlib lacks it: `minorDet` and its calculus, the expansion `det_mul_expansion`, and the rectangular Cauchy–Binet formula `cauchy_binet_rect` (not in Mathlib at this toolchain), yielding the compound-matrix identity `minorDet_mul`.
- **Task 2 — the dΓ lift: STANDS.** `SecondQuantDGamma.lean` (329 lines, 18 theorems, 3 definitions). `dΓ(M) = ∑ i, ∑ j, M i j • (a†ᵢ ∘ aⱼ)` as a finite sum of creation–annihilation composites. Proved: `dgamma_adjoint` (the adjoint of `dΓ(M)` is `dΓ(Mᴴ)`); `dgamma_one` (`dΓ(1)` is the Rung 8 number operator); sector preservation; `dgamma_basis_diag` — on a basis vector, `dΓ(diag λ) e_S = (∑ j ∈ S, λ j) • e_S`; and `dgamma_vacuum`.
- **Task 3 — the free Hamiltonian gap, DERIVED: STANDS.** For a diagonal one-particle Hamiltonian with eigenvalues `λ j ≥ 0`, set `freeHamiltonian λ = dΓ(diag λ)`. Proved: the vacuum is an eigenvector with eigenvalue `0` (`freeHamiltonian_vacuum`); every basis vector is an eigenvector with eigenvalue the sum of its occupied one-particle energies (`freeHamiltonian_basis`); self-adjointness; the expectation formula and its nonnegativity; and the derived gap itself — `freeHamiltonian_eigenvalue_gap`: if every one-particle eigenvalue is `0` or `≥ Δ`, then every non-vacuum eigenvalue of `H` is `≥ Δ`, with `free_gap_dichotomy` giving the eigenvalue dichotomy `0` or `≥ Δ`. **Label, precisely:** the one-particle gap `Δ` is an input, but the Fock-space gap is computed from the operator — as the sum of occupied one-particle energies — not assumed as a structure field and not inserted by definition. It is a statement about the **free** theory at **finite modes**. It is not the interacting Yang–Mills mass gap, and nothing in this rung touches the continuum theory.
- **Task 4 — quantum Poincaré implementation at finite modes: PARTIAL.** What lifts is proved, in form: `gamma_rep` — any one-particle unitary representation `ρ : G → Matrix` of any group `G` lifts through `Γ` to a unitary representation of `G` on Fock space (homomorphism, identity, vacuum fixed, sectors preserved, inner product preserved), discharging the form of Rung 6's second- quantization window at finite modes for any group whose action preserves the finite mode set. A concrete instance is implemented: `lattice_translation_rep`, the lattice translation group acting by diagonal phases, whose lift multiplies a basis vector by exactly the product of its occupied modes' phases — the finite-mode shadow of momentum additivity. What does **not** lift is priced exactly: (1) the mass-shell one-

particle representation (Rung 6's window stays T5; no Lorentz-invariant hyperboloid measure or massive one-particle Hilbert space exists in Mathlib to restrict to finitely many modes, and this rung does not construct one); (2) boosts and continuum rotations (a boost mixes any finite momentum set into infinitely many modes; no finite mode set is preserved, so there is nothing for Γ to lift — the liftable symmetry at finite modes is exactly the mode-set-preserving subgroup); (3) the infinite-mode / continuum limit (needs the infinite-mode Fock construction, priced since Rung 8); (4) interaction (Γ and dΓ are the free lifts; no interacting Hamiltonian, gauge field, or Yang–Mills dynamics appears in this rung).

The free fermionic theory at finite modes is now complete as a machine-checked package: Fock space (§8) + Γ + dΓ + free Hamiltonian with derived gap + a unitary symmetry representation in lifted form.

10. Rung 10 — Bosonic second quantization (T1) — STANDS (Tasks 1–3)

Two new modules (31 theorems and 13 definitions, zero sorrys; `BosonicFock.lean` and `BosonicAssembly.lean` unmodified). Labels that apply to everything below: **finite modes** (`Fin d`), operators on the **dense domain** `Dom d` (finite-particle vectors), **free bosonic theory**. This is not the interacting Yang–Mills theory and not a Millennium claim. The Task 2 gap is derived for the free Hamiltonian at finite modes — exactly the bosonic analog of §9 Task 3, with the same scope.

- **Task 1 — the bosonic dΓ lift: STANDS.** `BosonicSecondQuant.lean` (531 lines, 25 theorems, 5 definitions). `dgammaBos d M = ∑ i, ∑ j, (M i j) • ((adag d i) ∘ₗ (ann d j))` on `Dom d`, mirroring §9's fermionic `dgamma` with the Rung 8B domain adjoint relations in place of finite-dimensional adjoints. Proved: `dgammaBos_adjoint` (`⟪dΓ(M) x, y⟫ = ⟪x, dΓ(Mᴴ) y⟫` on `Dom d`); `dgammaBos_one` (`dΓ(1) = numberOp`, connecting to §8B's number operator); `dgammaBos_vacuum`; `dgammaBos_basis_diag` (`dΓ(diag λ) e_c = (∑ i, λ i * (c i : ℂ)) • e_c`); sector preservation (`dgammaBos_mem_domSector`), with `domSector_coe` coercing the domain sectors into §8B's Fock sectors.
- **Task 2 — the bosonic free Hamiltonian gap, DERIVED: STANDS.** For `freeHamiltonianBos d lam = dΓ(diag λ)` with `λ i : ℝ`, proved: every basis vector is an eigenvector with eigenvalue `∑ i, λ i * (c i)` (`freeHamiltonianBos_basis`, with the coefficient form `freeHamiltonianBos_coord`); the vacuum eigenvalue is `0`; self-adjointness on `Dom d`; the expectation formula (`freeHamiltonianBos_expectation`) and its nonnegativity for `λ ≥ 0`; and the derived gap — `free_gap_dichotomy_bos` and `freeHamiltonianBos_eigenvalue_gap`: every eigenvalue of `H` on a nonzero vector of `Dom d` is a basis eigenvalue, hence `0` or `≥ Δ` whenever each one-particle eigenvalue is `0` or `≥ Δ`. As in §9, the one-particle gap `Δ` is an input and the Fock-space gap is computed from the operator; the statement is about the **free** theory at **finite modes**.
- **Task 3 — bosonic Γ(U): PARTIAL at the rung's first landing; CLOSED by the Γ functor campaign (2026-10-16, later the same day).** `BosonicGamma.lean` (215 lines, 6 theorems, 8 definitions). The fermionic Γ of §9 used determinant

minors over subsets; bosons need **permanent** minors over occupation configurations, repetitions allowed. The lift is defined (`configsOfTot`, `configEnum`, `perMinor`, `normFactor`, `gammaKernel`, `gammaC`, `gammaBos`) and what the definition gives directly is proved: the kernel formula `gammaBos_apply_basis` (`Γ(M) e_c` is the sum, over configurations `c'` of the same total count, of `gammaKernel M c' c • e_{c'}`); `perMinor_of_tot_ne` (minors vanish across different total counts); `gammaBos_sector` (`Γ(M)` preserves the total-occupation sector); and `gammaBos_vacuum` (`Γ(M)` fixes the vacuum). **Priced remainder — the two obstructions, named exactly:** (1) `Γ(1) = 1` and the diagonal action `Γ(diag λ) e_c = (∏ i, λᵢ^{c i}) • e_c` both reduce, for this kernel, to the **fiber-permanent counting lemma**: the permanent of the fiber-indicator matrix (entries `1` where the enumerated positions of `c` agree, else `0`) equals `∏ i, (c i)!` — a finite but genuine counting argument about `Matrix.permanent` over repeated indices, not in Mathlib at this toolchain. (2) **Multiplicativity** `Γ(M)Γ(N) = Γ(MN)` and **unitarity** need a **permanent Cauchy–Binet identity for repeated indices**: expanding `per (M N)` over intermediate index functions and grouping them by their multiset image, with the `1/∏ c'!` weights matching the kernel normalization exactly — the bosonic analog of §9's rectangular Cauchy–Binet campaign, a separate campaign of comparable size. Until (1) lands, even `Γ(1) = 1` is unproved for this definition — so bosonic Γ is recorded as a candidate lift with its kernel formula, vacuum, and sector preservation proved, not as a verified functor. No statement was weakened to get here. **Closure (2026-10-16):** the functor campaign (`BosonicGammaFunctor.lean`, 1,161 lines, 42 theorems, 8 definitions, zero sorrys; targeted build green, 2,455 jobs; headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`) proved both obstructions with their consequences: the fiber-permanent counting lemma (`permanent_fiberIndicator_configEnum`), `Γ(1) = 1` (`gammaBos_one`), the diagonal action (`gammaBos_diagonal_basis`), the permanent Cauchy–Binet identity for repeated indices (`perMinor_mul`, with the multiplicity-weight bookkeeping proved), multiplicativity (`gammaBos_mul`), and the unitarity package (`gammaBos_adjoint`, `gammaBos_comp_conjTranspose`, `gammaBos_conjTranspose_comp`, `gammaBos_isometry`). **Task 3 is closed; bosonic Γ is a verified functor on finite modes, and Rung 10 stands whole.** Full record: `whitepaper/gamma-functor-campaign-2026-10-16.md`.

Priced remainder beyond this rung: the infinite-mode limit (no longer `Fin d`); the symmetric tensor-power unitary equivalence / permanent formula in Mathlib's tensor API (still blocked upstream, as priced at §8); and the standing frontier — continuum limit, interacting theories, Yang–Mills — which nothing in this rung touches.

10A. Pitch 2 — The free theory at countably many modes (T1) — STANDS WHOLE

The priced build order for the infinite-mode limit that §10 left open — scoped in `whitepaper/pitch-2-infinite-mode-scoping-DRAFT-2026-10-16.md` and executed in two campaigns (2026-10-16) — landed in full. The landing removes exactly one scope word from §§9–10: "finite modes" becomes "countably many modes." Everything else about the labels is unchanged: **free theory**, states on the **finite-particle domain**, and the

gap in the sense stated below. Five new modules, 137 theorem/lemma declarations and 57 definitions, zero sorrys; headline axiom checks (nine in Campaign A, eight in Campaign B) exactly `[propext, Classical.choice, Quot.sound]`.

- **Campaign A — the tower (57 declarations).** `FockLimit.lean` builds the shared machine Mathlib lacks: `onbEmb`, a linear-isometry constructor from a source Hilbert basis and a target orthonormal family, and density of finite-support vectors in the constant-ℂ `lp` spaces. `FermionicLimit.lean` and `BosonicLimit.lean` build the limit spaces — ℓ^2 over configurations on ℕ (Finset ℕ for fermions, $\mathbb{N} \to_0 \mathbb{N}$ for bosons) — with orthonormal configuration bases via `lp.single`, vacua, finite-particle domains and sectors; configuration embeddings between all finite levels and into the limit, with the cocycle proved in the **general** coherence form (all $d \le e \le f$, not merely successive steps) and the factorization that every limit configuration lives at a finite level. (A forced split, disclosed: `FermionicFock` and `BosonicFock` cannot share one Lean environment — both define `ThetLogos.adag`; the limit files inherit the split. The collision is the subject of the priced `adag` repair, §14.)
- **Campaign B — the operators and the gap (80 declarations).** `FermionicLimitOps.lean` and `BosonicLimitOps.lean`. The corpus Γ API is square-only, so a rectangular widening was ported (`minorDetRect`/`perMinorRect`, definitionally the square kernels) and the identifications proved at operator level: `gammaFermRect_pad` (whole-space operator equality, fermions) and `gammaBosRect_pad` (equality of domain maps — the honest bosonic landing, since bosonic Γ preserves domains, not whole spaces). The finite-particle domain of the limit is the union of the finite levels (`fermDomLim_eq_iSup_range`; `DomLim_eq_iSup_map` in domain-map form — the bare-range form is false for bosons, recorded), with density corollaries. Diagonal dΓ in the limit for both statistics: basis eigenvalue formulas, finite-level compatibility as operator equalities, and the number operator as the constant-1 case.
- **The gap, Δ-general (Task 6).** For a diagonal one-particle Hamiltonian whose eigenvalues are 0 or $\ge \Delta$ (any $\Delta > 0$), the free Hamiltonian in the limit has vacuum eigenvalue 0 and the eigenvalue dichotomy: every finite-particle eigenvalue is 0 or $\ge \Delta$ (`freeHamiltonianLimFerm_gap_of_orth`, `freeHamiltonianLimBos_gap_of_orth`), with self-adjointness on the domain. **Mandatory clause, carried in both module headers and repeated here verbatim in substance: this is the gap in the eigenvalue-on-finite-particle-domain sense of Rungs 9–10; the spectral statement (the gap as a statement about the full spectrum of the closed operator) is priced, unattempted** — the scoping's Task 7, a pre-agreed cut, as is Task 3b (rectangular functoriality beyond the padding identifications proved).

What Pitch 2 does not buy (its dossier's honesty line, adopted): not interaction — that is Pitch 3 (§10B), which is a separate construction at finite modes and has no limit-space form; not continuum or Poincaré (§13A); and not any part of any Millennium statement.

10B. Pitch 3 — Interaction at finite modes — STANDS WHOLE (both statistics; the Version 3 bosonic clause is discharged — §0, Corrections to Version 3)

The corpus's first interacting Hamiltonians. Both campaigns are finite-mode matter self-interactions — quartic Wick-ordered density-type terms added to the free Hamiltonians of §§9–10. No gauge field, no continuum, no Poincaré, and not the interacting Yang–Mills mass gap: that clause is carried in both module headers and applies to everything below. Scoping: `whitepaper/pitch-3-interaction-scoping-DRAFT-2026-10-16.md` (§6 named the landing shape and its failure consequence in advance; the verdict below applies §6 exactly as written).

- **Campaign 1 — fermionic interaction: STANDS WHOLE.** `FermionicInteraction.lean` (1,303 lines; 58 declarations, 15 definitions; build green, 3,297 jobs; headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The general interacting Hamiltonian `hamInt lam W g = freeHamiltonian + g • intOp W` is symmetric for every coupling tensor W, with g = 0 recovering the free Hamiltonian; the quartic `intOp W` is built on the corpus's first operator-norm bounds for the one-mode creators and annihilators, with every matrix coefficient and row sum of `intOp W` bounded by `intBound W = 2·∑‖W i j k l‖`. **Gap persistence (headline):** if `2·|g|·intBound W ≤ Δ/4`, there is a ground eigenvalue E_0 with `|E₀| ≤ |g|·intBound W`, and every eigenvalue is either in the ground cluster or ≥ Δ/2 — the free gap persists at size ≥ Δ/2 above the ground cluster, in the eigenvalue sense, at finite modes (`hamInt_gap_persistence`). The interaction is proved interacting: `densityInt_not_dgamma` — the density interaction equals `dΓ(M)` for no one-particle matrix M. Recorded in the module, per the house pattern: a scoped exact-gap statement under the weak dichotomy was **false** (two zero modes with U a b > 0 give an eigenvalue g·U a b ∈ (0, Δ)); the counterexample is in the theorem's doc-comment, the exact gap is proved under the strengthened no-zero-mode hypothesis, and the true weaker statements are proved instead.
- **Campaign 2 — bosonic interaction: STANDS except essential self-adjointness.** `BosonicInteraction.lean` (1,956 lines; 69 declarations, 21 definitions; targeted build green; headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). **B1 (Wick calculus):** monomial calculus on the finite-particle domain `Dom d` with the √-coefficient basis formula and number-relative bounds (`intBoundBos W = 2 · couplingMassBos W`); the symmetrized quartic `intOpBos W` is symmetric on `Dom d` for every W (`intOpBos_isSymmetric`). **B2 (self-adjointness, in three scoped senses):** symmetry of `hamIntBos lam W g` on the dense domain; sectorwise self-adjointness with full diagonalization (on every finite particle sector the operator is symmetric with an orthonormal eigenbasis and real eigenvalues — `hamIntBosSector_spectral`); and LinearPMap formal self-adjointness, `hamPMap ≤ hamPMap†` (`hamPMap_le_adjoint`). **B3(i) (exact diagonal gap):** under the **strong dichotomy** (`∀ i, Δ ≤ lam i` — no zero modes), every eigenvalue of the diagonal bosonic Hamiltonian is exactly 0 or ≥ Δ, uniformly over all sectors (`hamiltonianDensBos_gap_exact`). The weak-dichotomy form is **false**, by the same

two-zero-mode counterexample shape as Campaign 1, recorded in-doc; only its dominated form is proved. **B3(ii) (windowed persistence):** for an eigenvector in sector n, under the threshold `2 · (‖g‖ · intBoundBos W · (n+2)²) ≤ Δ/4`, the eigenvalue is within `‖g‖ · intBoundBos W · (n+2)²` of 0 or ≥ Δ/2 — the window is in the statement (`hamIntBos_gap_persistence_windowed`). The density interaction is again proved not to be a dΓ lift (`densityIntBos_not_dgamma`) — for both statistics now.

- **Disproved in-module (a result, recorded as one).** Uniform gap persistence for general bosonic quartic coupling is **false**: `densHamEigBos_unbounded_below` — attractive pair coupling (U a b = −1, g = 1) gives the eigenvalue (lam a + lam b)·n − n² at the (n, n) configuration, below every bound for large n. This proved collapse is the reason B3(ii) is windowed; it is also entered in the kill ledger (§15).
- **The one open clause — essential self-adjointness: PRICED at Version 3; DISCHARGED 2026-10-17 (ESA-α/β; §0, Corrections to Version 3, item 1).** The text of this bullet is Version 3's record of the clause as priced; the amendment paragraph after the verdict block below carries the discharge. What is not proved is the closure equality `hamPMap = hamPMap†` as partial linear maps. The obstruction, with one correction to the campaign dossier's account: the dossier stated that Mathlib has no LinearPMap closure operator; the parked-fleet gauntlet's audit (2026-10-17) found that stale — pinned Mathlib does define `LinearPMap.closure` (`Mathlib/Topology/Algebra/Module/LinearPMap.lean`), with `le_closure` and `IsClosable.closure_isClosed`, so the target is statable sharply as `hamPMap.closure = hamPMap†`. What remains absent is the analytic input: no Kato–Rellich theorem and no Nelson commutator theorem in the pinned Mathlib, and the closure's domain is an ℓ²-growth condition across infinitely many sectors. Two proof routes are named: R1, a Faris–Lavine/Nelson commutator argument with comparison operator (N+1)²; R2 (**recommended**), banded truncation — the quartic moves sectors by Δn ∈ {−4,−2,0,2,4} only, so the core property reduces to one ℓ² tail estimate. Re-priced **M–L** (restating the target: S).

  Pitch 3 verdict (the scoping §6's pre-named PARTIAL, narrowed to a single clause): Pitch 3 **stands for fermionic interaction whole**. For bosonic interaction, the Wick calculus with relative bounds, the exact diagonal gap, windowed persistence, sectorwise self-adjointness, and LinearPMap formal self-adjointness all stand; **essential self-adjointness is priced**. The §6 headline sentence ("…both statistics, with self-adjointness proved…") is **not** spendable in unqualified form, and no text in this paper spends it.

  Amendment (2026-10-18) — the verdict above is superseded in its one open clause. ESA-α (`BosonicESA.lean`, 17 declarations) proved the equivalence of the closure equality with the single inclusion `(hamPMap)† ≤ (hamPMap).closure` and the formal-adjointness direction; ESA-β (`BosonicESABeta.lean`, 16 declarations) proved the inclusion itself, by exact sector-preserving truncation (the balanced quartic preserves sectors exactly, Δn = 0, so an adjoint graph pair's occupation-cutoff truncations are exact graph pairs at every cutoff and converge to it), for **all** parameters with no hypotheses: `bosonicESA_proved`. **Pitch 3 stands whole at finite modes**

> **for both statistics, essential self-adjointness included.** The §6 headline sentence is spendable from this amendment forward — at finite modes, with the gap fences of this section (windowed persistence; uniform persistence disproved) unchanged. Full record: §0, Corrections to Version 3, item 1.

Route C of the scoping (interaction in the limit spaces of §10A) stays FRONTIER and is not attempted.

10C. Schrödinger dynamics — the evolution exhibit (T1) — STANDS

`SchrodingerDynamics.lean` (868 lines; 49 declarations, 10 definitions; build green, 3,319 jobs; ten headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). Finite dimensions only; the module takes no interpretation position (interpretations are T5), and whole-Fock-space/bosonic dynamics is FRONTIER and absent from it.

For any symmetric operator T on a finite-dimensional complex inner-product space, `evolOp hT t` is constructed diagonally — spectral phases in Mathlib's `eigenvectorBasis`. Proved: the group law (`evolOp_add`, with the inverse identities) and `evolOp_zero`; unitarity (`evolOp_isometry`; the linear isometry equivalence `evolUnitary` with inverse `evolOp (-t)`; `evolOp_adjoint`); the explicit solution formula (`evolOp_solution_formula`: coefficients evolve by eigenphases); **the Schrödinger equation as a derivative theorem** (`hasDerivAt_evolOp_apply`: dψ/dt = −i • Tψ) with **uniqueness** (`evolOp_unique`, by a coordinatewise integrating factor — no ODE library needed); eigenspace invariance. The Heisenberg side is built with its group law and algebra-automorphism package, and the Heisenberg equation (dA/dt = i[H, A] in the corpus's sign conventions). **Picture equivalence is a theorem** (`picture_equiv`): expectation values agree between the Schrödinger and Heisenberg pictures. **Headline application:** under the exact hypotheses of §10B's `hamInt_gap_persistence`, the ground cluster and the high (≥ Δ/2) cluster of the interacting fermionic Hamiltonian are subspaces, each invariant under the time evolution, covering the space with the ground cluster nontrivial (`groundCluster_invariant`, `highCluster_invariant`, `hamInt_clusters_cover_and_nontrivial`) — the gap persists not only under coupling but in time.

Two fenced shortfalls, named and priced: `evolOp` is characterized as the exponential by the Schrödinger equation plus uniqueness, but a literal identification `evolOp = NormedSpace.exp (-it•T̂)` was **not** proved (priced **S**); and the Dyson first-order term (interaction-picture, needs interval integrals of the Heisenberg-evolved interaction) was **not** landed (priced **S–M**). **Amendment (2026-10-18): both shortfalls are DISCHARGED** — `SchrodingerSmalls.lean` proves the literal exponential identification and the Duhamel/Dyson first-order package (§23); this section's remainders list is empty at its priced scope.

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11. SM-Match — structural match against the Standard Model (3 MATCH + 1 PARTIAL)

"Match" means: the formalized structures coincide with the Standard Model's known algebraic structure — same gauge group, same fermion quantum numbers, same classical Lagrangian shape, same breaking pattern. Masses, couplings, and mixing angles are **inputs** (labeled T2), never derived. This is a structural formalization result, not a new Standard Model and not a derivation of any measured value. 95 theorem/lemma declarations across four files (the program dossier's earlier estimate of "~70" understated the count; the verified figure is 95).

- **Track 1 — Full finite gauge algebra: PARTIAL.** On the algebraic side, the match is proved T1: the unitary group of $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ is U(1)×SU(2)×U(3) (`quat_mem_specialUnitary_iff`: $U(\mathbb{H}) = SU(2)$), and its unimodular subgroup is exactly U(1)×SU(2)×SU(3) (`SMUnimodular_mem_iff`). On the represented side, the Rung 2 caveat is located exactly, not redefined: the kernel of the Option-A representation is characterized (`pi_ker_iff`); $\mathbb{C}$ and $\mathbb{H}$ survive faithfully; $M_3(\mathbb{C})$ survives only locked to the electroweak sector; pure color dies. And the priced gap is proved, not asserted: `pi_u1_eL_trivial` — the represented U(1) acts trivially on the left electron, so **the represented U(1) is not the Standard Model hypercharge U(1)**. A faithful representation of the full SM gauge group on the finite Hilbert space is not achieved; the gap is priced at the pre-Option-A representation or the manifold factor.
- **Track 2 — Fermion content: MATCH.** The eight fermion types carry the Standard Model's (Y, T^3, color) assignments as labeled T2 inputs (hypercharges are not derived; generation triplication is imposed, per `ThreeGen.lean`). Proved T1: electric-charge consistency $Q = T^3 + Y/2$ for all eight types (`charge_consistency`), and consistency of the index↔fermion mapping with the finite-triple embedding. Priced: full SU(3) color is unrepresented in the Option-A representation (Track 1's gap).
- **Track 3 — Classical Lagrangian sectors: MATCH.** 19 declarations: three-factor gauge kinetic term with joint gauge invariance proved (`smGaugeKinetic_gauge_invariant`), the correct $-¼$ sign, and nonnegativity; the fermion term's fluctuation split (`fermionTerm_split`); the Higgs potential's minima structure (pure algebra); the Yukawa sector structure with an input matrix and no values. Priced: the quantum Lagrangian, the continuum integral, the junk quotient, and all coupling values.
- **Track 4 — Symmetry breaking: MATCH.** 42 declarations. The VEV stabilizer is exactly span(Q) (`higgs_stabilizer_is_Q`) — the precise algebraic statement of $SU(2)_L \times U(1)_Y \to U(1)_{em}$; the VEV minimizes the potential; the photon direction is massless because Q is unbroken (`higgs_photon_direction_massless`); $M_W = gv/2$ and $M_Z = \sqrt{(g^2+g'^2)}v/2$ hold as formal relations among the input parameters (g, g′, v), with no numerical values anywhere.

12. Theories intake, Round 1 — fit verdicts

Standing rule: a theory is admitted only if its core structures map onto the program's formalized objects with real theorems. Refusals are results.

- **KMS / Tomita–Takesaki: FIT at finite level.** 10 T1 theorems in `KMSBridge.lean`, including the finite-dimensional KMS identity for Gibbs states (`gibbs_kms_weight`) and the equivalence of KMS equilibrium with vanishing thermal flux; two previously pinned T5 targets in the modular-flux file are discharged. Priced T5 window: unbounded Tomita–Takesaki theory and type-III factors — Mathlib has no von Neumann algebra theory.
- **Classical Yang–Mills: FIT classically; quantum NO-FIT.** 16 T1 theorems in `ClassicalYangMills.lean`, including the represented curvature from 1-form data with exact gauge covariance (`repCurvature_gauge_covariant`) and gauge invariance of the Tr(F†F) action. The quantum theory — quantization, continuum, mass gap — is recorded NO-FIT and remains quarantined T5.
- **AQFT / Haag–Kastler: per-axiom.** In `HaagKastlerFree.lean` (14 declarations): isotony FIT (T1); locality FIT at creator level and for fermionic even observables (T1), with the honest caveat that the spacelike hypothesis is vacuous at creator level; Poincaré covariance PARTIAL (geometric half T1, implementer priced); vacuum PARTIAL (cyclic vector T1, invariance priced); spectrum condition and irreducibility NO-FIT as theorems in the current setting (priced windows recorded).
- **Spectral action / finite index bridge: FIT at finite level (2026-10-11 verification); build failure found and repaired 2026-10-15 — see the correction below.** On the formalized `smDirac`: Tr(1) = 32; Tr(D_F^2) as the explicit Yukawa-sum formula; the fluctuated trace expansion; vanishing of odd traces; the finite-dimensional Fredholm index and the chiral-compression index (zero). NO-FIT recorded: the continuum heat-kernel expansion and the Atiyah–Singer theorem (both need continuum elliptic analysis). **Correction (2026-10-15):** re-verification for this paper found that `SpectralActionFinite.lean` no longer compiles under the pinned toolchain — four proof failures (`simp` made no progress / failed rewrite at lines 942, 1006, 1024, 1074), all inside the a_2 trace-computation proofs and consistent with Mathlib deprecation drift (`if_pos`/`if_neg` simp behavior). `SpectralActionBridge.lean` fails transitively. The theorem statements are unchanged and no `sorry` was introduced, but until the proofs are repaired this entry's T1 status is **not currently machine-verified** and the FIT verdict is suspended, not standing. Repair is a queued, well-localized job. **Resolution (2026-10-15):** the repair was completed the same day — proof-local fixes only (re-applying the diagonal multiplication reduction inside the affected trace computations); no statement changed; no `sorry`/`admit`/axiom added; the file builds green again and **the FIT verdict stands restored** (theories-intake dossier amended; repair commit `6b4e2f9`).

Round 2 queue (not yet run): BRST/anomalies; the Osterwalder–Schrader reconstruction bridge; noncommutative integration / Dixmier trace.

Note (2026-10-17). Round 2 has still not been run as an intake. Its Osterwalder–Schrader item has meanwhile split (per the parked-fleet gauntlet's ruling): the fit-audit half is killed as redundant — Pitch 4's Stage 1b (§13A) is now the program's OS work, done as construction rather than audit — while the re-pinning half stands: the corpus's `OsterwalderSchrader.lean` header premise ("Mathlib has no tempered distributions") is stale (pinned Mathlib has $\mathscr{S}$ and the Fourier transform; the true absences are the smearing map, the full field φ(f), and OS verification), and the reconstruction dependency audit remains priced S–M. BRST stands priced M; the Dixmier trace item stands priced S–M.

Note (2026-10-18) — Round 2 is CLOSED. The re-pin landed (§25): the stale premise is discharged on genuine $\mathscr{S}$, the old reconstruction and existence statements are proved vacuous and re-pinned with a `NonGaussian` nontriviality clause, and reconstruction is pinned as a schema whose two analytic holes (Bargmann–Hall–Wightman continuation; edge-of-the-wedge) are named and absent from the pinned Mathlib. The Dixmier intake landed the same day (§25): the definition pins as data, the log-mean algebra and measurability criterion stand, the finite triple's Dixmier trace is **identically zero** — the gauntlet's predicted finite shadow, tested and wrong, corrected in the open — and the harmonic profile carries trace exactly its coefficient under every generalized limit. The BRST intake also landed (§25): finite Chevalley–Eilenberg cohomology with $d_2 \circ d_1 = 0$ proved as the Jacobi contraction, and $\mathfrak{su}(2)$'s $H^0 = \mathbb{C}$, $H^1 = 0$, $H^2 = 0$ computed exactly — FIT as algebra, NO-FIT as quantization.

13. The Morley edge-signature gauntlet — PROMISING, with one precise kill

Hypothesis (Jeff): a 4D boundary carrying two 2D edge operators generates the 3+1 split, with the torsion/twist in the D_F projection carried, not assumed away. Steelmanned, then formalized in `EdgeSignature.lean` (30 theorem/lemma declarations; the gauntlet dossier groups the core results into four sections). Build status: verified green by the gauntlet on 2026-10-11; the 2026-10-15 group rebuild timed out without errors, and the file was **re-verified later on 2026-10-15 by a standalone build (exit 0; zero `sorry`/`admit`; axiom checks on the 13 headline theorems clean)** — see §0. The verdicts below stand as the record.

- **Morley core — T1.** Edge-pair ⟺ fundamental symmetry: from $\eta^2 = 1$, the operators $E\pm = (1\pm\eta)/2$ are idempotent, orthogonal, and complementary (`edgePair_of_symm`); conversely a perfect pair gives $\eta = E_+ - E_-$ with $\eta^2 = 1$ (`symm_of_edgePair`). The "arbitrary input → perfect structure" pattern has real formal content: perfection is forced by $\eta^2 = 1$ alone.
- **Flip twist carried — T1.** The doubled edge decomposition is flip-invariant for (η, η): the Rung 2 twist preserves the perfection.
- **Torsion carried exactly — T1.** $(\eta+T)^2 = 1 + (\eta T + T\eta + T^2)$ (`torsion_defect`); fluctuated edge operators miss idempotence by exactly $(2^{-1})^2 \bullet (\eta T + T\eta + T^2)$ (`edgePlus_torsion_defect`); perfection survives torsion iff $\eta T + T\eta + T^2 = 0$, a

Maurer–Cartan-type flatness condition (`torsion_flat_iff`). (The earlier $\mathbb{C}^7$-torsion claim remains killed and is not resurrected.)

- **Strong emergence — KILL, by T1 counterexample.** `signature_not_determined`: (1,1) and (1,−1) on $\mathbb{C}\times\mathbb{C}$ both square to 1, but one has vanishing minus-edge (signature (2,0)) and the other does not (signature (1,1)). $\eta^2 = 1$ does not determine the signature; the 3+1 numbers are extra data. The kill is itself a theorem.
- **Analogy assessment:** real formal content at the algebraic level; the "miracle" grading (maximal symmetry from no symmetry) remains philosophy, labeled NEEDS-WORK as an analogy.

Priced next steps: a D_F-specific torsion map (η built from Dirac data, defect computed under inner fluctuations); the dimensional reading — "two 2D" suggests neutral signature (2,2), and what breaks it to (3,1) is open; integration with the twisted first-order condition; a boundary spectral triple as the formal home of the "4D boundary."

13A. Pitch 4 — The continuum passage, opened on the lattice — Stages 0, 1a, 1b stand as stated; NO CLAIM on the pitch

Pitch 4 is the program's scoped route from its finite constructions toward the continuum Yang–Mills theory. Its scoping (`whitepaper/pitch-4-continuum-scoping-DRAFT-2026-10-17.md`) is chartered **NO CLAIM**: before every stage below, and after all of them, **parts of the official Clay statement closed: 0**. The scoping's decisions, stated once here: the route is **lattice first** — the analytic route (continuum fields via Minlos-type measure constructions) is priced XL with no landing, blocked at its first move by the absence of nuclear spaces and a Minlos theorem from the pinned Mathlib. The stage map: Stage 0, finite lattice gauge theory; Stage 1a, correlations; Stage 1b, reflection positivity and its consequences; Stage 1c, strong-coupling clustering (priced XL, separately killable); Stage 2, one exact coarse-graining step (M–L; the iterated flow with estimates is XL, Balaban-scale); Stages 3a/3b, OS-reconstruction scaffolding (M / M–L); Stage 3c, uniform estimates along a continuum trajectory — **priced research-scale and open in print**, with one audit rule the scoping states plainly: deriving the continuum gap from a gap assumed uniform along the trajectory is circular as an audit instrument and is forbidden in this program's writing; Stage 4, an interacting Poincaré representation in the continuum — FRONTIER. Two traps are named in advance: Elitzur's theorem (gauge-variant quantities cannot carry expectations — met, and discharged, at Stage 1a) and triviality (the φ^4_4 precedent: a lattice route can end in a free continuum theory; nothing below speaks to that question).

All stages below are finite-lattice T1. The lattice is not the continuum, no limit of any kind is taken or claimed in any of them, and none of them is a step in a Clay proof — they are the first priced bricks of the road the scoping draws.

- **Stage 0 — finite lattice gauge theory: STANDS WHOLE.** `LatticeGauge.lean` (1,656 lines; 68 declarations by the §0 strict method — 72 counting its four `@[simp]` loop lemmas, which the strict rule excludes; 33 definitions and instances; build green, 3,390 jobs). Sites form an abstract finite type Λ with four shift

bijections, concretely the torus `Fin 4 → ZMod N`; links are group-valued, `U : Link Λ → G` for a general compact group G; the plaquette and Wilson action are built from a general unitary representation ρ. Proved: **exact gauge invariance of the Wilson action for every gauge transformation** — no gauge fixing anywhere — with invariance of the product-Haar configuration measure in the strong form (`Measure.map (gaugeAction σ g) (configMeasure μG) = configMeasure μG`); Wilson loops gauge-invariant; the normalized Haar probability `haarProb` built in-file from Mathlib's Haar measure, its right- and inversion-invariance proved from uniqueness. The **β = 0 package at general (G, ρ)**: at zero coupling the Gibbs expectation is the product-Haar expectation, the single-plaquette expectation factorizes through the Haar intertwiner P, and `gibbsExpect_zero_plaquette : ⟨W_p⟩₀ = tr P` — which vanishes exactly when ρ has no nonzero fixed vectors (`gibbsExpect_zero_plaquette_eq_zero`), with the circle and full-unitary instances discharged on the torus. Action nonnegativity for β ≥ 0.

- **Stage 1a — correlations and Elitzur: STANDS WHOLE** (pitch-level shape after this stage: PARTIAL-β). `LatticeCorrelations.lean` (1,667 lines; 68 declarations, 13 definitions; build green, 3,391 jobs; the five scoped headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The Gibbs measure is constructed as an actual probability measure at **every real β** (`gibbsMeasure`, with `isProbabilityMeasure_gibbsMeasure`), and Stage 0's ratio-of-integrals expectation is proved equal to its integral (`gibbsExpect_eq_integral`) — the measure Stages 1b–1c consume. The expectation algebra (constants, additivity, scaling, finite sums, $|\langle F\rangle| \le \langle|F|\rangle$, the real form with monotonicity) and the covariance package: symmetry, the centered form, **variance nonnegativity** (`gibbsVar_nonneg`), and **Cauchy–Schwarz** (`gibbsCov_sq_le`, by the elementary discriminant argument). **Elitzur's theorem at general (G, ρ), for every β** (headline): the mechanism theorem `expectMatrix_eq_zero_of_left_covariant` — the expectation matrix of any left-covariant insertion is zero whenever ρ has no nonzero fixed vectors — with corollaries `elitzur_single_link` and `elitzur_open_wilson_line` and the torus instances. The pre-authorized narrowing to instance-level Elitzur was **not** needed. The contrast is proved, not gestured: plaquette expectation matrices are intertwiners, not vanishing subjects (`expectMatrix_plaquette_intertwines`). β = 0 factorization for disjointly supported observables landed **stronger than scoped** — the identities carry no integrability hypotheses at all — and is fenced to β = 0 in the module, its dossier, and here. The Wilson two-point function is defined as an object, with the bound $|\langle W_1 W_2\rangle| \le d^2$ and the connected function defined; **no decay or clustering claim is made at any β** (that is Stage 1c, untouched).
- **Stage 1b-A — reflection structure: STANDS WHOLE.** `LatticeReflection.lean` (1,335 lines; 67 declarations, 16 definitions; build green, 3,392 jobs). The abstract-Λ presentation of Stages 0–1a provably carries no reflection (a grep-verified obstruction recorded by the gauntlet), so the reflection is built on the concrete torus: the **between-sites** reflection in direction 0, `reflSite x = update x 0 (-1 - x 0)` — an involution with the shift calculus $\theta\sigma_0\theta = \sigma_0^{-1}$, $\theta\sigma_i = \sigma_i\theta$ ($i \neq 0$), and **no fixed sites at even N**. The fixed link stratum is **characterized by an if-and-only-if** (`reflLink_eq_self_iff`): the fixed links are exactly the time links based

in the two layers adjacent to the two reflection planes — not enumerated, characterized; the pre-named enumeration-only PARTIAL shape was not triggered. The configuration reflection factors as θ = invConfig ∘ permConfig, and **both the product-Haar measure and the Gibbs measure are reflection-invariant at every real β**, in the strong `Measure.map` form (`gibbsMeasure_map_reflConfig`) — stronger than the pre-named Haar-level fallback. **The Wilson action split is proved term by term**: $S = S_+ + S_- + S_0$, with $S_+(\theta U) = S_-(U)$, $S_-(\theta U) = S_+(U)$, and S_0 — the crossing class of temporal plaquettes over the two fixed layers — θ-invariant term by term.

- **Stage 1b-B — reflection positivity: PARTIAL (the pre-named shape γ, extended).** `LatticePositivity.lean` (1,559 lines; 44 declarations, 10 definitions; build green, 3,393 jobs). The observable class is the **strict positive half** `posLinks = halfLinks \ planeLinks`; the exclusion of the reflection-fixed plane links is forced — two finite counterexamples are recorded in the stage dossier (hand computations, not Lean theorems): a plane-link observable over $G = \mathbb{Z}/3$ at $\beta = 0$ gives the form value -2, and at fixed stratum variables a U(1) character gives $I_1(\beta)e^{i\gamma}$, not even real. What stands, at general (G, ρ) and **every real β**: the OS form `osForm` (θ on observables = conjugation ∘ the observable ∘ the configuration reflection), its **conjugate symmetry** (`osForm_conj_symm`, via 1b-A's Gibbs reflection invariance), reality of the diagonal, full sesquilinearity, the bridge to the bare-pullback reading of $\langle\theta(F)\cdot\bar{F}\rangle$ (`osForm_bare_eq`), and the **kernel identity** (`osForm_eq_kernel`): the form is Z^{-1} times a double integral over the two halves, with the crossing action in the kernel. **Positivity is proved at β = 0** (`osForm_zero_nonneg`: the weight is 1, the kernel factors, and the diagonal is $W\cdot\bar{W}$). The **Gram criterion is proved, β-sign-free** (`osForm_nonneg_of_gram`): if the Gibbs weight in split variables is a finite Gram sum $\sum_i \mathrm{conj}(\Xi_i\circ\mathrm{mirror})\cdot\Xi_i\circ\mathrm{restrict}$ with bounded measurable features, then the diagonal is $Z^{-1}\cdot\sum_i\|\int f\cdot\Xi_i\|^2 \geq 0$. Positivity at general β is thereby reduced, losslessly, to **one named open clause**: the Gram (character-expansion) factorization of the Wilson weight. Its standard proof needs Schur/Peter–Weyl orthogonality of matrix coefficients on a general compact group, which the pinned Mathlib (v4.35.0-rc3) does not contain — its representation theory covers finite groups and the algebraic theory. **Price: XL, blocked on Mathlib infrastructure.** The positivity theorem itself is classical (Osterwalder–Schrader; Glimm–Jaffe); it is unproved here, **not** killed.
- **Stage 1b-C — the OS Hilbert space: STANDS.** `LatticeOSSpace.lean` (1,065 lines; 42 declarations, 17 definitions; build green, 3,550 jobs; eleven headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). Built by the **time-slice L^2 route** (the gauntlet's ruling; the quotient/completion-of-the-form route was declined): `osSpace μG ρ β := Lp ℂ 2 (sliceMeasure μG ρ β)`, where the slice measure is the Gibbs marginal on the time-0 slice — a probability measure at **every real β**, and at β = 0 exactly the product Haar measure over the slice. The space is a Hilbert space by Mathlib's Lp instances — positive by construction, with **no appeal to OS positivity**. The **vacuum** is the constant function 1, of norm 1 (`norm_osVacuum`); vacuum expectation equals the Gibbs expectation of the pulled-back observable (`vacuum_expectation`). Bounded slice observables act as multiplication operators (`sliceMul`), operator norm ≤ the bound, with **adjoint =**

pointwise conjugation and real-valued ⇒ self-adjoint. What routes through 1b-B's open clause is stated **conditionally on it, by name**: the clause is packaged as `GramClause`, and `osForm_nonneg_of_gramClause` / `osForm_slice_nonneg_of_gramClause` assume it as an explicit hypothesis; positivity at β = 0 for slice observables is unconditional (`osForm_slice_zero`). One marginal identity (the pushforward of product Haar along the slice inclusion) is named, unproved, priced **S**, and **not used**.

- **Stage 1b-D — the transfer matrix: STANDS.** `LatticeTransfer.lean` (1,610 lines; 58 declarations, 13 definitions; build green, 3,551 jobs; eleven headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The transfer operator is the adjoint compression `T_β = A*† ∘L B*` on the OS space, where A* (`slicePull`) pulls a slice observable back along the time-0 restriction and B* (`nextPull`) does the same along the time-1 restriction transported to the time-0 slice; both legs are linear isometries into the same L^2 of the Gibbs measure, and both slice marginals are proved equal to the slice measure at every real β (translation invariance of the Gibbs measure, proved in the module in `Measure.map` form, by 1b-A's reflection argument run for translations). The operator package, at **every real β**: the **form identity** (`transferOp_inner`: ⟪g, T_β f⟫ is the two-slice Gibbs kernel pairing, as an identity); **contraction** (`transferOp_norm_le`: ‖T_β‖ ≤ 1, by Cauchy–Schwarz against the two isometries); **vacuum fixed** (`transferOp_vacuum`: T_β Ω = Ω); **self-adjointness** (`transferOp_adjoint`, `transferOp_isSelfAdjoint`, at even N — from 1b-A's proved reflection invariance via the reflection step Ψ = (time shift by −2) ∘ θ, which exchanges the two slices; **no clause spent**); and **form positivity** (`transferOp_form_nonneg`: for pointwise real nonnegative slice observables, ⟪g, T_β f⟫ is real and nonnegative — proved from the positivity of the Gibbs measure alone, again with no appeal to OS-form positivity or the Gram clause). **The lattice gap at β = 0 is exact:** the kernel pairing factorizes over the disjoint slices (Stage 1a's β = 0 factorization), giving `transferOp_zero_apply : T₀ f = ⟪Ω, f⟫ • Ω` — **T_0 is exactly the orthogonal projection onto the vacuum line**, idempotent (`transferOp_zero_comp_self`), killing the vacuum-orthogonal complement; its spectrum is {1} on the vacuum line and {0} on the complement, so **the gap of T_0 is exactly 1**. That is the free (infinite-temperature) point; it is exact, and it is all the gap this stage claims.

Pitch-level shape after Stage 1b. Stages 0 and 1a stand whole; Stage 1b stands in A (structure), C (the space), and D (the operator package with the exact β = 0 gap), and in B it stands in the form, at β = 0, and as the proved Gram criterion, with general-β positivity the stage's single open clause. **PARTIAL-δ** — the scoping shape that would let this paper say "reflection positivity stands" for Stage 1b — is **NOT claimed**, and will not be claimable until 1b-B's clause closes. Stages 1c, 2, 3a/3b, 3c, and 4 are unbuilt and priced (§14). Nothing in this section takes a thermodynamic or continuum limit, constructs an interacting Poincaré representation, or closes any part of the Clay statement. **Amendment (2026-10-18): Stage 2 and Stage 1c-0 have since landed (§21)** — Stage 2 whole (one exact blocking step; the qualitative constraint set of this section proved closed under it; no quantitative uniform bound manufactured), Stage 1c-0 PARTIAL (β-analyticity, the first-order perimeter law, and rigorous $O(\beta^2)$ stand; the fully general intertwiner closed form for κ_p was cut above its price and is named, not

hidden). PARTIAL-δ remains **NOT claimed**; Stages 1c proper, 3a/3b, 3c, and 4 remain unbuilt and priced (§14).

14. Frontier and priced windows

The following are named, framed, and priced — not claimed.

1. **Unbounded operator theory in Mathlib.** Densely defined closed operators, the unbounded adjoint, von Neumann's A** theorem, the Cayley transform, projection-valued measures, the unbounded spectral theorem, and Borel functional calculus are absent from Mathlib (surveyed 2026-10-11; five checkable window statements are filed in `SpectralFrontier.lean`). This is the boundary of formalized analysis on the spectral prong.
2. **Second quantization (bosonic) — discharged in Version 2.** The adjoint-relation assembly is discharged (§8B), and Γ(U) functoriality is discharged (§10, Task 3, closed by the Γ functor campaign of 2026-10-16): `Γ(1) = 1`, the diagonal action, multiplicativity, and unitarity are proved at finite modes. The **infinite-mode limit is now also discharged** (Version 3): the direct-limit Fock construction exists and the free theory stands at countably many modes (§10A, Pitch 2). What remains priced on this line: the **spectral** form of the free gap at countable modes (the eigenvalue-on-finite-particle-domain form is proved; Task 7 of the Pitch 2 scoping), rectangular functoriality beyond the padding identifications (Task 3b), interaction in the limit spaces (§10B's Route C: FRONTIER), and the symmetric tensor-power unitary equivalence (blocked upstream, as priced at §8).
3. **Boost invariance** of the mass-shell measure — **DISCHARGED in Version 3** (§7 amendment): the one-dimensional Jacobian identity is proved and, with the factorization cap, the measure is invariant under the full restricted Lorentz group.
4. **Strong continuity** of the one-particle representation: **DISCHARGED in Version 4 (§22)** — translations by the named dominated-convergence route (dominator $4\|\psi\|^2$) and the Lorentz subgroup for the full restricted Lorentz group, at the identity and at every point. **Fock packaging: DISCHARGED** by the concrete orthonormal-basis route (§22; the abstract route stays killed, §15), and Fock-level Lorentz strong continuity with it, for both statistics. What remains priced on this line is the one statement no brick has attempted: strong continuity of the joint action in a topology on the full Poincaré group — every result here is subgroup-wise, and no joint topology on `RPoincare` exists in the corpus (unpriced; §22).
5. **n-point Wick expansion:** the Wick recursion at 2n points.
6. **Faithful SM gauge representation** (§11, Track 1): priced at the pre-Option-A representation or the manifold factor.
7. **Continuum limit and interacting 4D QFT.** No construction exists in this corpus. The Wightman verdicts of Rung 5 are toy-model verdicts; the general theory is not formalized.

8. **Yang–Mills existence and mass gap.** Quarantined T5. The Clay statement is formalized as a fixed target (`ClayStatement.lean`), and the dossier discipline records why finite-level gaps do not transfer: Osterwalder–Seiler's strong-coupling lattice gap excludes the continuum limit; Balaban's renormalization-group work is not a mass-gap proof; φ^4_4 triviality says nothing directly about Yang–Mills; spectral-gap undecidability results concern families of Hamiltonians, not the single Yang–Mills Hamiltonian.
9. **Bosonic essential self-adjointness** (§10B): **DISCHARGED in Version 4** (§0, Corrections to Version 3, item 1; ESA-α/β) — by neither priced route: exact sector-preserving truncation made the domain argument exact at every cutoff. The Yukawa Hamiltonian's essential self-adjointness (§18) is a separate, still-priced instance of the same shape.
10. **The `adag` repair and Yukawa: DISCHARGED in Version 4.** The repair landed (2026-10-17), and the Yukawa campaign on top of it landed with it (§18) — its fenced walls met exactly as priced: its own essential self-adjointness not proved (item 9's discharge is for the Pitch 3 Hamiltonian, not this one), and persistence windowed only, uniform persistence false by §10B's counterexample embedded at the fermion vacuum. The repair program then found and discharged two further fences of the same kind (`numberOp`; the packaging six), so no pair of corpus modules is fenced from co-import (§26).
11. **Reflection positivity at general β** (§13A, Stage 1b-B): the Gram/character-expansion factorization of the Wilson weight. Priced **XL, Mathlib-blocked** (needs Peter–Weyl orthogonality for general compact groups). Until it closes, PARTIAL-δ stays unclaimed.
12. **The lattice gap at general β** (§13A, Stage 1b-D): spectral input on the transfer operator at fixed $\beta > 0$ (classically, compactness of the continuous-kernel operator and its spectrum). Priced **M–L**. A gap uniform in volume and coupling along a continuum trajectory is item 13's Stage 3c, not this.
13. **Pitch 4's stages** (§13A, §21): Stage 1c-0 **landed PARTIAL in Version 4** (§21: analyticity, the first-order perimeter law, and rigorous $O(\beta^2)$ stand; the fully general intertwiner closed form for the first-order plaquette coefficient was cut above M and is named — any reading of the exhibit as area-law evidence remains killed in advance); Stage 2, one exact blocking step, **landed whole in Version 4** (§21; the iterated flow with estimates stays XL, Balaban-scale). Still unbuilt: Stage 1c proper, uniform clustering (XL, killable — any clustering statement routing through gauge-variant intermediates is dead on arrival by §13A's Elitzur theorem); Stages 3a/3b, OS scaffolding (M / M–L — their continuum-side target is now pinned, §25); Stage 3c, uniform estimates along the continuum trajectory (research-scale, open in print — **the missing ingredient**, named as such by Stage 2's closure probe: the qualitative constraint set is proved closed under blocking, and no quantitative uniform bound exists in it to close); Stage 4, interacting Poincaré (FRONTIER).
14. **Schrödinger smalls** (§10C): **DISCHARGED in Version 4** (§23) — the literal `NormedSpace.exp` identification and the full Duhamel package with integral

remainder both stand. No scattering theory and no Dyson-series convergence is claimed there or anywhere.

15. **Bridge follow-ons** (Appendix A): **DISCHARGED in Version 4** (§23) — attainment of the Fiedler infimum (with no connectivity hypothesis), the minimizer's eigenvector property with the λ_2-identification, exponential decay at the Fiedler rate, and the Faraday storey, which stands fenced as graph bookkeeping (discrete Stokes, `d² = 0`, EMF as a definition-theorem pair), **not** Maxwell theory.

16. **Decoherence toy: DISCHARGED in Version 4** (§24) — the absence was re-verified (grep, corpus and pinned Mathlib) and the partial trace built from scratch, with the pinching toy on it.

17. **Reconstruction's analytic core** (§§25, 27): the Bargmann–Hall–Wightman analytic continuation and the edge-of-the-wedge theorem — absent from the pinned Mathlib (grep-verified 0 hits), and the one remaining hole in the now-fully-concrete OS reconstruction schema. Research-scale. Until it exists, no OS or Wightman reconstruction theorem is statable in this corpus beyond the schema, and none is claimed.

18. **The resonance spectrum's general-generator reduction** (§24): the phase law is proved for diagonal generators; reduction of a general Hermitian generator to its eigenbasis form is named, not attempted (the thermal instance stands because the corpus's clocks are already eigenbasis-packaged).
19. **The thermal pre-compensation instance** (§24): pre-distorting a state so that Gibbs thermalization — acting through the clock's flow, not a Schur-type family — lands it on target. Named remainder of the pre-compensation brick; unpriced.
20. **Open Problem 2** (§20): a tripotent triple generating all twelve `smGen` from witnesses outside the three named sector families — in particular from non-$\mathbb{Z}[i]$-valued tripotents, which the screening wave's entry-ring invariant cannot see — or a class-closing invariant for the unrestricted problem. UNPRICED (research). The sector sharpening is refuted (§20); the problem is open.

15. Kill ledger (selected)

Killed claims return only as new entries by proof; the originals stand.

- As-stated Connes–Kreimer antipode axiom — refuted; corrected version proved (different claim).
- ρ_M ⊗ ρ_F as product twist — does not typecheck.
- Strong Morley emergence (edge data determine 3+1) — T1 counterexample.
- p/n as a fundamental gauge-coupled ± pair; electron as an interaction product of that sector — killed by standard physics (isospin is an approximate global symmetry; lepton number, spin, and statistics all fail for the composite-electron reading). The honest survivors: approximate global isospin, the electron as a separate lepton field, beta decay as weak charged current.

- W5 monomial-range cyclicity as framed — false; span version proved.
- Bosonic-form W6 criterion for fermions — fermion-blind; Koszul version proved.
- "Finite triple converges" (continuum language) — killed; priced roadmap instead.
- The φ-tower (golden-ratio mass hierarchy) — killed against measured mass ratios.
- $\mathbb{C}^7$ torsion — killed; not resurrected by the Morley gauntlet.
- Uniform bosonic gap persistence for general quartic coupling — **disproved in-module** (§10B): attractive coupling gives eigenvalues (lam a + lam b)·n − n², unbounded below. The windowed statement is the true one and is proved.
- The weak-dichotomy exact-gap forms of both Pitch 3 campaigns — false by the two-zero-mode counterexample recorded in both modules; the strengthened (no-zero-mode) statements are proved as different claims.
- Positivity of the Osterwalder–Schrader form over observables supported on the reflection-fixed plane links — false (Stage 1b-B dossier's counterexamples: form value −2 at β = 0 over $\mathbb{Z}/3$; a U(1) character value $I_1(\beta)e^{i\gamma}$, not real, at fixed stratum variables). The strict-positive-half class is the true one.
- The abstract SymmetricAlgebra route to Fock packaging — killed (again, and permanently, in the 2026-10-17 gauntlet); the concrete orthonormal-basis port is the only route (§14, item 4).
- A separate Osterwalder–Schrader fit-audit intake (the Round 2 item) — killed as redundant: absorbed by Pitch 4's construction (§12 note). Its re-pinning half survives as a priced item, not as an audit.
- The XYZ sector grid — all 24 named triples (`genC`, `genH k`, `genM a`) killed (21 by the $\mathbb{Z}[i]$ entry-ring invariant, 3 at the tripotence screen), and with them the sector existence statement `TriadicSectorExistence` itself: **refuted** (§20). Open Problem 2, which quantifies over all matrices, is a different statement and remains open (§14, item 20).
- The 2026 OS and Wightman existence statements as shapes — proved vacuous (inhabited by the all-`True` instantiation), and Rung 4's `SharpWightmanTarget` as an existence form — proved vacuous by junk data (§27). Withdrawn as statements; the re-pinned failable forms stand in their place (§§25, 27). The six sharp definitions as conditions are not killed and not impugned.
- The Dixmier intake's predicted finite shadow ("the noncommutative integral as a finite residue/trace functional") — tested and **wrong**: the finite Dixmier trace is identically zero, proved (§25). The functional in the finite noncommutative-integral slot is the ordinary trace.
- The trit direction (ternary representation as a separable energy lever for inference, and with it the trit framing of the data-center question) — **DIES-K3** by its own pre-registered kill: the modeled 58% advantage was bit-width, not base (§28).

16. What is NOT claimed (the immune system)

- **No 4D QFT.** The finite triple is zero-dimensional; the Wightman verdicts are one-mode toy verdicts; no interacting theory in four dimensions, or in any continuum, is constructed. (Interacting Hamiltonians at finite modes are constructed — §10B; the fence is stated at "No interacting continuum theory" below.)
- **No continuum limit.** The continuum is a priced roadmap throughout.
- **No mass gap.** The Clay target is fixed and quarantined; nothing here is a step in a Clay proof, only infrastructure such a proof would need. Parts of the official Clay Yang–Mills statement closed by this corpus: **0** — before and after every result in this paper.
- **No full SM gauge group from the finite representation.** The represented U(1) is provably not hypercharge (§11). The algebraic match holds at the level of the full finite algebra and its unimodular subgroup.
- **No derived masses, couplings, or mixing angles.** All are labeled inputs wherever they appear.
- **No physical predictions.** Spectral, thermal, and trace numbers in this corpus are properties of the mathematical models, not measurements of nature.
- **No derivation from analysis where axiomatization was used.** Rung 3's promotions and Rung 4's bridge axiom are labeled T2 at every citation.
- **The free-theory gaps are not the mass gap.** The derived gaps of §9 (Task 3), §10 (Task 2), and §10A (Pitch 2, at countably many modes) are statements about free Hamiltonians, with the one-particle gap Δ as an input; the persistence result of §10B is a statement about a finite-mode interacting Hamiltonian under its threshold; and the lattice gap of §13A is a statement about a finite-lattice transfer matrix at $\beta = 0$. None of them is the Yang–Mills mass gap, and none transfers to any interacting or continuum theory (§14, item 8).
- **No interacting continuum theory.** (Amended in Version 3; the Version 2 line "no interacting theory of either statistics" is overtaken by §10B and is withdrawn.) Interacting Hamiltonians at finite modes are constructed for both statistics (§10B) — matter self-interaction only. (Amended in Version 4:) a mixed fermion–boson interaction of Yukawa **shape** is also constructed at finite modes (§18) — the `adag` collision that prevented even stating it is discharged (§26). It is not the Standard Model Yukawa sector: no gauge field, no Higgs mechanism, no flavor or chirality structure. No gauge-coupled interaction appears anywhere in this paper. No scattering content, no interacting Poincaré representation, and no interacting theory in the continuum is constructed or claimed.
- **No continuum Γ.** (Amended in Version 3; the Version 2 line "No infinite-mode or continuum Γ" was false from 2026-10-16 and is withdrawn.) Γ is verified at finite modes (§§9–10) and, for the free theory, at countably many modes (§10A). No continuum lift of Γ is proved or claimed.
- **No interacting Poincaré representation, and no joint full-group continuity.** (Amended in Version 4; the Version 3 line "No Poincaré representation on Fock space" is overtaken by §22 and is withdrawn.) What is now proved: the free mass-

shell measure is invariant under the full restricted Lorentz group (§7 amendment); the one-particle representation is strongly continuous for both subgroups (§22); and its Γ-port to both Fock spaces is a unitary group action with translation and Lorentz strong continuity (§22). What is not proved: strong continuity of the joint action in a topology on the full Poincaré group (no joint topology is introduced anywhere), any boost or rotation as a strongly continuous representation of an interacting theory, and the interacting continuum Poincaré representation — Pitch 4's Stage 4, FRONTIER, unbuilt.

- **No reflection positivity at general coupling.** Stage 1b-B (§13A) proves the OS form, its positivity at $\beta = 0$, and the Gram criterion; positivity at general β is reduced to one named open clause (the Gram factorization of the Wilson weight, XL/Mathlib-blocked). **PARTIAL-δ is not claimed.** Nothing in §13A's Stages 1b-C/1b-D spends that clause: the OS space is positive by construction, and the transfer matrix's proved properties route through 1b-A's proved reflection invariance and the Gibbs measure's positivity alone.
- **No lattice gap at general coupling, and no uniformity.** The transfer matrix's gap is proved exactly at $\beta = 0$ only (§13A). At general β the gap is fenced and priced (§14, item 12); no gap uniform in volume or coupling is stated anywhere, and no continuum-limit conclusion is drawn from any finite-lattice statement in this paper (Stage 3c's uniform estimates are research-scale and open in print; Stage 4 is FRONTIER).
- **Self-adjointness is claimed exactly where it is proved, and nowhere else.** (Amended in Version 4.) For the Pitch 3 bosonic Hamiltonian, essential self-adjointness is now proved (§10B, as amended; §0, Corrections to Version 3), so the unqualified statement stands for that operator, at finite modes. For the Yukawa Hamiltonian (§18) only the scoped senses stand — symmetry on the dense product domain, LinearPMap formal self-adjointness, and bandwise spectral theory of the free part — and its essential self-adjointness is **not** proved and **not** claimed. No other operator in this corpus is called self-adjoint beyond the senses its section states.
- **No bridge from the background algebra.** The modules of Appendix A (the $\mathbb{Z}_3$ and $\mathbb{Z}_{60}$ gradings, the flux-twist toy, the weld, and the graph-Laplacian bridge exhibit) are T1 algebra standing alone: no $\mathbb{Z}_3$ grading of the TRO, of any tripotent family, or of any Dirac operator is claimed, and no physical content is asserted for them. The bridge exhibit's own completion (§23) changes nothing here: its graph-Faraday package is bookkeeping on a finite graph, not Maxwell theory, and no bridge from it to §13A's lattice or to any field theory is claimed.

17. Reproducibility

- **Source:** https://github.com/jeffgurd888/thet-logos (Lean sources in `lean/ThetLogos/`; verdict dossiers in `whitepaper/`).
- **Build:** with Lean 4 / Mathlib per `lean/lean-toolchain` and `lean/lake-manifest.json`: `cd lean && lake build` (green; covers the root-imported modules, and was green at 3,325 jobs at the Rung 9, Rung 8B, and Rung 10 landings). The remaining

modules build individually: `lake build ThetLogos.<Module>` — the 2026-10-15 status per module is reported in §0, the Version 2 modules' targeted builds are reported in the §0 addendum, and the Version 3 modules' targeted builds (each green, with job counts) are reported in the §0 Version 3 addendum. The Version 3 inventory was taken at commit `4380a2a`.

- **Counts:** the counting method of §0 (Version 1–2 numbers) and of the §0 Version 3 addendum (all Version 3 and Version 4 numbers, one uniform method) reproduce every number in this paper from the sources. The Version 4 inventory was taken at commit `3447e09`. The program total reproduces, from the repository root, with the one strict recount: strip Lean comments from every file in `lean/ThetLogos/` and count lines matching `^(theorem|lemma)` — the program chain of the §0 addenda sums those per-module counts to 3,276 (the whole-tree count, 3,945, includes the non-chain exploratory files and is not the program total; both figures are stated in the §0 Version 4 addendum so the distinction can be checked, not trusted).
- **The sampler, re-run.** The vacuum-portrait exhibit (§28) is a numerical reproducibility exhibit in the same spirit: its sampler runs from a fixed seed (20261018), and an independent re-run of the full pipeline reproduced every reported number and all six rendered panels byte-identically. Samples are not theorems; the exhibit's fences are printed with it (§28).
- **Axioms:** `#print axioms <theorem>` in any module reproduces the footprints reported in §0.

18. The Yukawa interaction — the first mixed fermion–boson term (T1) — STANDS, in windowed form only

`YukawaInteraction.lean` (2,355 lines; 119 declarations, 44 definitions; build green, 3,313 jobs; five headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The corpus's first interaction mixing the two statistics — made statable by the `adag` repair (§26). Labels that apply to everything below: **finite modes only** (df fermionic modes, db bosonic modes); **Yukawa shape only**; **not a mass gap**.

- **The space and the operator (STANDS).** The joint space is `lp` over product configurations `Finset (Fin df) × Config db` — deliberately not a completed tensor product: the two factors share one orthonormal basis, as in both parent campaigns. The interaction is `H_Y = Σ_{i,j,k} (Y_{ijk} + conj Y_{jik}) • (a†ᵢ aⱼ) ⊗ (a†ₖ + aₖ)` — a fermionic density/current bilinear tensored with a bosonic field — and the total Hamiltonian is `H = H_free + (g : ℂ) • H_Y`. Cross-factor commutation is a **theorem** (four identities on the nose), not a convention; the coupled monomial's matrix element factors (density element × field element).
- **A kernel-caught fence: the coupling is real.** Symmetry of `H` on the product domain holds with `g : ℝ`; for complex `g` the two inner-product slots produce `conj g` versus `g` and the operator is not symmetric. The module takes `g` real throughout, and the reason is recorded in it.
- **Self-adjointness — scoped senses ONLY (STANDS as scoped).** `H` is symmetric on the dense product domain, with `hamYukPMap ≤ hamYukPMap†` (LinearPMap formal

self-adjointness); the free Hamiltonian preserves every bosonic band and is diagonalized there by an exhibited orthonormal eigenbasis. The interaction does **not** preserve bands (it maps band N into band N + 1, proved), so no bandwise spectral theorem for `H` itself is stated. **Essential self-adjointness of `H` is NOT proved and NOT claimed** — Pitch 3's ESA discharge (§10B) is for the Pitch 3 Hamiltonian, not this one; the priced remainder carries over verbatim.

- **Gap persistence — WINDOWED ONLY (STANDS in that form).** With per-mode gap Δ on both free factors (strong dichotomy), an eigenvector of `H` supported in bosonic band N at the explicit threshold `2 · (‖g‖ · yukBound Y · √(N+1)) ≤ Δ/4`, with `yukBound Y = 4·Σ‖Y i j k‖`, has eigenvalue within `‖g‖ · yukBound Y · √(N+1)` of 0 or ≥ Δ/2. **No uniform persistence is claimed: it is false in general for bosonic factors** — §10B's collapse counterexample embeds at the fermion vacuum sector. The joint vacuum is an exact 0-eigenvector of `H` at every coupling, and `H_Y` is proved not to be a diagonal (dΓ-type) operator whenever some `Re (Y i i k) ≠ 0`.

Fences. This is a matter coupling of Yukawa shape (fermion bilinear × boson field). It is **not** the Standard Model Yukawa sector: no Higgs mechanism, no flavor or chirality structure is present or claimed. No gauge field appears. No infinite-mode or continuum limit is taken.

19. The thermal clock — every faithful state carries its clock (T1) — STANDS

`ThermalClock.lean` (33 declarations after the housekeeping closure below, 12 definitions; build green, 3,310 jobs; nine headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`, a footprint that also certifies the quarantine below). Finite level only; the clock is a property of a **state**, not of spacetime; the Connes–Rovelli thermal-time hypothesis remains T4/T5 literature, neither proved nor assumed here.

- **The definition (pinned, derived).** `ThermalClock N` packages a faithful finite state by its eigenbasis weights (positivity and normalization as fields); the state ρ, the modular Hamiltonian `K = −ln ρ`, and the flow are derived from the weights, not stipulated. `K_apply_diag` proves K's diagonal entries are exactly `−log` of ρ's; K is Hermitian. The matrix-picture companion `GibbsClock N` carries the temperature field β > 0 with generator β·K.
- **Additivity, in all three pictures.** Flowing s + t is flowing t then s — proved in the eigenbasis picture (`diagFlow_add`), the matrix picture (`modularFlow_add`), and as transport of the corpus's modular-data group law. A change of weight normalization does not change the clock (`diagFlow_scale_weights`).
- **The conversion (headline): for Gibbs states, physical time = β × thermal time.** Proved in three forms: the eigenvalue form (`gibbs_K_eigenvalue_diff`: the generator's level spacings are `K_i − K_j = β·(λ_i² − λ_j²)`, the logZ constant cancelling in the open); the eigenbasis flow form (`diagFlow (gibbs weights) s = hamiltonianFlow (λ²) (β·s)`, as maps on all observables, with the clock proved independent of the logZ convention outright); and the basis-free matrix form

(`modularFlow (β·K) s = modularFlow K (β·s)`). The rate β is constant — the Gibbs clock ticks uniformly; at β = 1 the clocks coincide.

- **The stopped clock is a proved case, not an embarrassment.** The tracial state ρ = I/N has identically trivial flow (`maximallyMixedClock_stopped`), as does any constant-weight clock; any observable commuting with the state is fixed by its clock; and at equilibrium the Gibbs clock's flux vanishes (`gibbs_flux_vanishing`) — its hands exert no force. The Gibbs state is the KMS equilibrium state of its own clock (`GibbsClock.kms`).
- **Quarantine, machine-checked.** Nothing in the module routes through the corpus's one T3 `sorry` (`ModularTime.kms_identity_diagonal`): the conversion is assembled from the eigenvalue-difference anchor and direct phase algebra instead, and the axiom checks above — `sorryAx` absent from all nine headlines — are the proof of that statement.
- **The scalar-shift closure (housekeeping, 2026-10-18).** The dossier's one named leftover — the standalone matrix-level lemma that the flow is blind to a scalar shift of the generator — is now proved, appended as the module's §7 (+4 declarations): `[K + c•I, A] = [K, A]` for every A; the modular flux is unchanged by `K ↦ K + c•I`; **the flow is unchanged** (`modularFlow_add_smul_one`); and the conversion therefore holds for the **full** Gibbs generator `β•K + (ln Z)•I`, not only its non-scalar part (`gibbs_conversion_full_generator`). The ThermalClock shelf's named leftovers are empty. (Route note, recorded in the dossier: the exponential of the shifted generator splits by `Matrix.exp_add_of_commute` and the scalar exponential is evaluated by `Matrix.exp_diagonal`; the originally sketched algebraMap route needs a `NormedRing (Matrix)` instance the pinned Mathlib does not provide.)

20. The triadic pin and the XYZ screening wave — the sector grid REFUTED; Open Problem 2 OPEN

The heat/gravity/light triad of the XYZ gauntlet received its pin on 2026-10-17 and its screening wave on 2026-10-18. Both landings are reported here at their exact tiers.

- **The pin (definitions only; `TriadicPin.lean`, 4 definitions, no theorems claimed by the pin itself).** The gauntlet's premise on the gravity leg was **overturned** in the pinning: the a_4 Yukawa evaluation it priced as open is proved in the corpus (`tr_DF_fourth_yukawa`); what remains open on that leg is the fluctuated a_4, and the Einstein–Hilbert identification is a dictionary (T2), with the continuum reading NO-FIT. The light leg is PARTIAL: no quantized electromagnetic field exists at either corpus address, and the represented-U(1) caveat of §11 stands (`pi_u1_eL_trivial`). The pin's negative result is its sharpest: **the assignment of the three gauntlet "principle seats" (heat/gravity/light) to the corpus's sector families is nomenclature (T2), not a derivation.** The pin's target statement is `TriadicSectorExistence`: tripotents `X ∈ sectorC`, `Y ∈ sectorH`, `Z ∈ sectorM`, one from each named family, whose generated TRO contains all twelve `smGen`.

- **The wave (`TriadicScreening.lean`, 283 lines; 19 declarations, 1 definition; build green, 3,129 jobs; every theorem's axioms exactly the standard three — in particular no `sorryAx`: nothing depends on the quarantined `xyz_tripotents_exist`).** Tripotence of the M family is now determined for **every** member: `genM 0`–`genM 6` are all tripotents (1–6 by this wave, via one uniform theorem), and `genM 7` is not (prior corpus). The grid has 24 triples. **All 24 are DEAD — ALIVE 0, UNRESOLVED 0.** Twenty-one die by the entry-ring invariant R5 (for `a ≠ 7` the three members are ℤ[i]-valued, ternary products preserve the entry ring, and the generated TRO provably misses `smGen 11`); the three with `a = 7` die at the tripotence screen. Assembled, the 24 kills prove ¬ `TriadicSectorExistence` (T1): the pin's sector sharpening is **REFUTED**. The kill is reported as a result, at the same speed as a positive landing.
- **The fence, exact: Open Problem 2 remains OPEN.** `xyz_tripotents_exist` quantifies over all matrices. A witness built from tripotents outside the three named families — in particular from non-ℤ[i]-valued tripotents, which R5 cannot see — is untouched by every screen here; the corpus's quarantined `sorry` at `XYZTripotents.lean:492` is neither used nor discharged. The wave also records why no invariant of the R5 family can close the unrestricted problem (an entry-ring invariant obstructs only triples valued in that ring), and prices the unrestricted problem — a construction, or a class-closing invariant — UNPRICED (research) (§14, item 20). Per the house relabeling rule, any future candidate enters as a new candidate, not as a resurrection of the refuted sharpening.

21. Pitch 4 continued — Stage 2 STANDS; Stage 1c-0 PARTIAL

Both stages are finite-lattice T1. Neither takes a thermodynamic or continuum limit; neither claims a gap at general β; **PARTIAL-δ is still NOT claimed**; parts of the Clay statement closed: 0.

- **Stage 2 — one exact blocking step: STANDS WHOLE.** `LatticeBlocking.lean` (1,654 lines; 60 declarations, 15 definitions and 1 structure; build green, 3,558 jobs; 13 headline axiom checks exactly the standard three). One exact factor-2 blocking step, built abstract-first: a `BlockingScheme` (an injective coarse-site embedding with the two parity axioms), with the torus (`2M → M`, coordinate doubling) as its instance; blocked links are path-ordered products of the two fine links; the blocked measure is the pushforward. Proved: blocked variables are gauge-covariant for the canonical lift of the coarse gauge action; the blocked Gibbs measure is gauge-invariant in the strong `Measure.map` form at **every real β** and is a probability measure; and at β = 0 the pushforward of the product Haar measure **is** the coarse product Haar measure — so the blocked β = 0 theory equals the coarse β = 0 theory, and the whole Stage 0/1a β = 0 package (intertwiner expectations, the vanishing criterion, disjoint-support factorization, the blocked plaquette expectation `tr P`) transports verbatim. Two blockings compose, as functions and as measures. The blocked measure's **exact effective action** exists at every β — the two measures are mutually absolutely continuous, and the blocked measure is exactly the coarse configuration measure weighted by `exp(−S_eff)` with `S_eff` the minus-log of the Radon–Nikodym density. Whether

`S_eff` is of single-plaquette Wilson form at β > 0 is **not** claimed: it is fenced as the named proposition `BlockedGibbsIsCoarseGibbs`, proved at β = 0 and open at β > 0 — a Stage 3 question.

- **The constraint-closure probe (the stage's strategic result).** Asked whether the proved constraints of Stages 0–1b survive blocking verbatim, the module answers item by item, all proved: gauge invariance at every β — **yes**; the β = 0 theory itself — **yes, verbatim** (it is equal, not merely equivalent); Elitzur vanishing for blocked links at every β — **yes** (with a torus instance); Wilson loops — **yes** (a coarse loop equals the fine loop over the doubled path). And the set's edge is proved, not gestured: a fine single-link observable does **not** factor through the blocking map (`singleLink_not_factor_blockMap`) — fine single-link data is exactly what blocking integrates out. The reading, labeled as the dossier's interpretation: the constraint set {gauge invariance, β = 0 exactness, Elitzur, Wilson-loop factorization} is **proved closed under one exact blocking step**. Every member of that set is a qualitative (exact, structural) constraint. **Blocking manufactures no quantitative uniform bound**: no constraint in Stages 0–1b bounds a gap or a correlation length uniformly, and Stage 3c's uniform estimate remains **the missing ingredient**, at its research-scale price (§14, item 13).
- **Stage 1c-0 — the β-analyticity exhibit: PARTIAL.** `LatticeAnalyticity.lean` (2,505 lines; 87 declarations; build green, 3,530 jobs; nine headline axiom checks exactly the standard three). **Fixed finite volume only** — every constant below is volume-dependent and displayed as such. What stands: β-analyticity of Gibbs expectations at 0, via an explicit power series with explicit radius `(2·max(actionBound, 1))⁻¹`, `actionBound = 2·(16·|Λ|)`; the first-order expansion with a closed, explicit O(β²) constant (`‖⟨F⟩_β − (⟨F⟩₀ − β·Cov₀(F, A))‖ ≤ firstOrderConst(‖F‖)·β²` for |β| ≤ 1), with the derivative identity `d/dβ ⟨F⟩_β|₀ = −Cov₀(F, A)`; the link-sharing reduction at general (G, ρ) — `Cov₀(F, A)` is an exact finite sum over plaquettes sharing a link with F's support, every other term killed by Stage 1a's β = 0 factorization; and on the torus the **perimeter law at first order**: `‖Cov₀(W(C), A)‖ ≤ 48·d·|∂C|` — the first-order response of a Wilson loop sees its perimeter, its link count, linearly. The narrowing, named: the first-order plaquette coefficient κ_p stands at general (G, ρ) as that exact finite link-sharing sum, and its scalar closed evaluation stands on the circle instance — **κ_p = −1/2**, hence `⟨W_p⟩_β = β/2 + O(β²)` (for G = Circle, fundamental representation, torus N ≥ 2; the mixed moments vanish by a witness-link case analysis on character orthogonality proved from translation invariance alone). The **fully general closed intertwiner form** for arbitrary (G, ρ) — via the J-tensor / ρ⊗ρ̄ pair-intertwiner apparatus — was **cut**: priced above the exhibit's M mid-scoping, and recorded here as cut, not as failed and not as proved. This is the gauntlet's pre-authorized landing shape for the exhibit. Kill, restated in advance and kept: **nothing in this exhibit is area-law evidence.** The perimeter law is a statement about the first-order response at β = 0, where only boundary links can appear; area behaviour is a β > 0, all-orders question, neither proved nor suggested here. No decay or clustering claim is made at any β. (Consistency note, not a proof obligation: the vacuum-portrait sampler of §28 measured ⟨Re U_p⟩ ≈ 0.12404 at β = 0.25 on a 6^4 torus, against this section's first-order prediction β/2 = 0.125.)

22. Strong continuity, completed — translations, Lorentz, Fock packaging, Fock-level Lorentz (T1)

This section discharges, in four landings, the strong-continuity and packaging remainders that §7 and §14 carried since Rung 7. All four are **free-theory** statements. Every continuity result is **subgroup-wise**: no joint topology on the full Poincaré group is introduced anywhere (§16, as amended).

- **Translations (one-particle): STANDS.** `TranslationContinuity.lean` (220 lines; 10 declarations; build green, 3,393 jobs; headline axioms exactly the standard three). For every mass m > 0 and every ψ ∈ L² of the mass shell, ‖U₁(a, 1)ψ − ψ‖₂ → 0 as a → 0 — the exact statement of the priced window `StrongContinuity_window`, proved by the priced route: translations act as pure phase multiplication (the shell action is trivial for ⟨a, 1⟩, a pointwise identity), so the squared difference integrand is dominated by 4‖ψ‖² and filter-form dominated convergence closes it. The group law transports continuity at 0 to every point.
- **Lorentz (one-particle): STANDS WHOLE.** `LorentzContinuity.lean` (28 declarations; build green, 3,661 jobs; headline axioms exactly the standard three). For every m > 0 and every Ψ ∈ L² of the shell, the packaged Lorentz action Λ ↦ U₁⟨0, Λ⟩ Ψ is continuous at the identity and at every Λ₀ — for the **full restricted Lorentz group**, boosts and rotations together, no axis or subgroup restriction; the priced PARTIAL fallback was not needed. A content point, stated plainly: the corpus had **no topology on the Lorentz group** anywhere; the module equips it with the topology of pointwise convergence of the action — equivalently (proved) entrywise matrix convergence, which in finite dimensions is the usual Lie-group topology — and continuity is proved for that topology. The route is the coefficient trick: transport to ℝ³ by the graph parametrization, one change of variables moves Λ onto the fixed factor of the cross term (so no dominator for the moved factor is ever needed — the translations dominator provably does not generalize), dominated convergence on the dense class of compactly supported continuous functions, and the ε/3 extension by isometry.
- **Fock packaging: STANDS-AS-PRICED.** `FockPackaging.lean` (38 declarations), `FockPackagingBos.lean` (99), `FockPackagingFerm.lean` (78) — **215 declarations**; targeted builds green (3,675 jobs across the brick); headline axioms exactly the standard three. The one-particle action is packaged on a **concrete** orthonormal basis `shellONB` of the shell space (normalized Gaussian-polynomial patches transported along the shell parametrization; separability and infinite-dimensionality come from the concreteness), and ported through explicit limit kernels in that basis — normalized permanents (bosonic) and determinant minors (fermionic) of the one-particle entry matrix — to unitary group actions `gammaFockU1` / `gammaFockFU1` on the assembled bosonic and fermionic limit Fock spaces. Proved in both statistics: the representation law (in the corrected applied order, Γ(g₁·g₂) = Γ(g₂) ⋙ Γ(g₁) — a correction to the window's display under Mathlib's `trans` convention, recorded in the dossier, not a narrowing), the identity law, unitarity on the closure of every n-particle sector, vacuum invariance, and **translation strong continuity transported** through finite-particle density. **Kill K4 stands**

permanently: the abstract SymmetricAlgebra route to this port is dead (§15) and was not used; no abstract Γ functor is invoked anywhere in the brick.

- **Fock-level Lorentz: STANDS WHOLE, both statistics.** `FockLorentzBos.lean` and `FockLorentzFerm.lean` (7 declarations each; builds green, 3,672 and 3,669 jobs; headline axioms exactly the standard three). For every Fock vector Ψ, `Λ ↦ Γ(U₁⟨0, Λ⟩) Ψ` is continuous at the identity and at every Λ_0, in both statistics — the transport the one-particle brick named as its fenced remainder. The route replays the packaging's translation chain with the one-particle Lorentz input swapped in: the packaging bricks' kernel lemmas were already filter-general, so **no tensor power of the one-particle space is ever formed and no new tensor lemma was needed** — in this packaging the Γ kernel in the concrete basis is the finite tensor data (recorded because the step had been priced as the brick's possible real content). The free theory's strong-continuity window is now closed at every level the corpus packages: translations and Lorentz, one-particle and Fock — subgroup-wise, as fenced above.

23. Three remainders, discharged — Schrödinger smalls; the Bridge completion

- **Schrödinger smalls: STANDS WHOLE.** `SchrodingerSmalls.lean` (487 lines; 24 declarations, 5 definitions, zero `@[simp]`; build green, 3,497 jobs; seven headline axiom checks exactly the standard three). Both of §10C's priced shortfalls, at their priced scope, finite dimensions only. **(a) The literal exponential:** `evolOp_apply_eq_exp` — `evolOp hT t ψ₀ = NormedSpace.exp (t • genCLM hT) ψ₀`, where `genCLM hT` is the generator `(−i)•T` packaged as a continuous linear map; proved because the exponential curve satisfies the same Schrödinger equation with the same initial value, and §10C's uniqueness theorem identifies them. (One form note: in the pinned Mathlib, `NormedSpace.exp` is a single field-independent function, so the ℝ/ℂ bridge the scoping budgeted friction for does not arise.) **(b) The Dyson first-order term and Duhamel:** the interaction-picture perturbation `V_I` for Pitch 3's fermionic `hamInt`, packaged as a CLM curve; `W₁(t) = (−i)•∫₀ᵗ V_I(s) ds` with its FTC derivative theorem; and the **exact** Duhamel identity for the interaction-picture state, `F(t)ψ₀ = ψ₀ + W₁(t)ψ₀ + ∫₀ᵗ −i•V_I(s)(F(s)ψ₀ − ψ₀) ds` — the remainder is a theorem-level integral term, so "first-order" here is an approximation statement, not a notation, with its Schrödinger-picture form. **Fences:** the Dyson term is a formal first-order coefficient plus an exact integral remainder — **no scattering theory, no S-matrix, no Dyson-series convergence claim**; fermionic environment only. §10C's remainders list is empty at its priced scope.
- **Bridge completion: STANDS WHOLE, including Faraday as bookkeeping.** `BridgeCompletion.lean` (1,194 lines; 60 declarations, zero `@[simp]`; build green, 3,316 jobs; nine headline axiom checks exactly the standard three). `BridgeLaplacian.lean` is unmodified; this module discharges its three priced follow-ons. **Attainment:** the Fiedler infimum of Appendix A is attained by an admissible potential — with **no connectivity hypothesis** (if the graph is disconnected in the support of w then fiedler w = 0 and every statement remains true verbatim). **The**

λ₂-identification: the minimizer is an eigenvector of L with eigenvalue fiedler w, and every mean-zero eigenvector has eigenvalue ≥ fiedler w — so **fiedler w IS the smallest eigenvalue of L on the mean-zero subspace**, an attained eigenvalue and a floor for the others. **Heat decay:** for the finite heat semigroup defined eigenbasis-wise, `‖F(t) − mean‖ ≤ exp(−(fiedler w)·t)·‖F₀ − mean‖` for t ≥ 0, with the derivative theorem (the flow solves Ḟ = −LF). **Faraday, fenced:** the storey is built Bridge-locally (the gauntlet ruled §13A's lattice does not cheapen it — group- valued holonomies and additive real drops differ in kind): faces as ordered vertex quadruples, flux through a face defined as its boundary circulation, the discrete Stokes identity (the flux of a potential field through any face is zero — the graph's `d² = 0`), and `inducedEMF` with its derivative theorem (EMF = −dΦ/dt as a definition-theorem pair). It stands as **graph-Faraday bookkeeping on a finite weighted graph — not Maxwell theory**; no coupled induced-field dynamics is built or claimed.

24. Channels — the decoherence toy, pre-compensation, and the resonance spectrum (T1)

Three finite-level bricks about what channels and flows do to states. All are finite-dimensional; none is a claim about any physical system beyond its stated finite model.

- **The decoherence toy: STANDS at its priced scope.** `DecoherenceToy.lean` (581 lines; 37 declarations, 6 definitions; build green, 3,244 jobs; 13 headline axiom checks exactly the standard three). **TOY, by charter.** Its first deliverable is infrastructural: the corpus's **first partial trace** — `partialTraceB`, defined by block trace — built from scratch after the absence was re-verified by grep over the corpus and the pinned Mathlib (0 hits in both). Proved for it: trace preservation; positivity preservation (in the corpus's quadratic-form clause of `IsDensityMatrix`, through the embedding identity — the reduced quadratic form at v is the sum over environment slices of the joint form at the embedded vectors); the product rule `Tr_B(A ⊗ₖ B) = Tr(B)·A`; star compatibility; the trace-duality characterization that pins the map uniquely; and **reduced states of density matrices are density matrices**. On it, the dephasing toy: the channel as diagonal pinching in a fixed product basis, with populations and trace invariant, purity `Tr ρ²` non-increasing with equality **iff the state was already diagonal**, and pinching commuting with the partial trace — so after one joint pinching every off-diagonal of the reduced state is exactly zero. One priced wording is corrected by the toy itself: the priced "geometric decay" of coherences degenerates — off-diagonals do not shrink by a factor per step, they are **exactly zero after ONE application** and stay zero (`pinch^[k] = pinch` for k ≥ 1); the priced statement is proved in the stronger exact form. No environment dynamics, no interaction Hamiltonian, no rates or timescales; the measurement problem is not touched in either direction.
- **Pre-compensation: STANDS whole at its priced scope.** `PreCompensation.lean` (28 declarations, 4 definitions; build green, 3,245 jobs; nine headline axiom checks exactly the standard three). Origin, recorded in the dossier: gear teeth are machined deliberately wrong — tip relief, crowning, twist compensation — so the

loaded gear runs true; the transplant is to prepare a state pre-deformed so a known channel lands it exactly on target. The channel family is **partial dephasing** `dephase l A = l•A + (1−l)•pinch A` (diagonal fixed, off-diagonals scaled by l; l = 1 the identity, l = 0 the toy's pinching), proved to map density matrices to density matrices for 0 ≤ l ≤ 1, with levels composing multiplicatively (`dephase a ∘ dephase b = dephase (a·b)`). **The reachability theorem (headline):** for l ≠ 0, a target ρ is exactly pre-compensable **iff** the de-amplified candidate `dephase l⁻¹ ρ` is itself a density matrix — the candidate is the unique possible pre-image, and of its three density conditions Hermiticity and trace are automatic, so reachability is exactly the candidate's positivity. At l = 1 every target is reachable; **at l = 0 the reachable targets are exactly the diagonal ones** — the channel's forgetfulness as a proved obstruction; diagonal targets are reachable at every level. For 0 < l < 1 the reachable set is a **proper subset, proved**: the maximally coherent pair state (rank one; populations ½, ½; coherence ½) is not pre-compensable at any such l — its de-amplified candidate has quadratic form exactly `1 − l⁻¹ < 0` on `eᵢ − eⱼ`. In plain terms: undoing partial dephasing means amplifying coherences by 1/l before the channel shrinks them back, and amplification is legal only while the amplified state is still a state — the tooth can only be pre-cut so far. **The gear case proper:** for a known unitary error, every target is exactly pre-compensable, by inverse preparation (`unitary_precompensable`) — systematic deformation cancels completely, because conjugation forgets nothing; the contrast with dephasing is the theorem's content. **Fences:** fluctuating (stochastic) noise admits no exact pre-image trick, and no theorem here is about fluctuations; `dephase l` outside [0, 1] is a linear map, not a claimed channel; the **thermal instance** — pre-distort so that Gibbs thermalization, acting through the clock's flow rather than a Schur family, lands on target — did **not** fall out of this machinery and is named as the remainder (§14, item 19), not papered over.

- **The resonance spectrum: STANDS, with one named narrowing.** `ResonanceSpectrum.lean` (15 declarations, 2 definitions; build green, 3,317 jobs; all 15 headlines' axioms exactly the standard three — nothing routes through the quarantined `ModularTime` sorry). The pin gives an ontological sentence two load-bearing clauses and proves each at finite level. **(I) Organisation = the operator's fixed-point set:** for every dephasing level l ≠ 1, the fixed points of `dephase l` are exactly the diagonal matrices (`dephase_fixed_iff_diagonal`) — the whole family shares the pinching's fixed-point set, so the organised states are the eigenvalue-1 space of the environment's channel at every nonzero forgetting strength. **(II) Resonance = the gap set of the generator:** for the diagonal-generator phase flow (proved definitionally the corpus's `hamiltonianFlow` on `Fin N`), the **phase law** `(flow t A) i j = exp(−it(λᵢ − λⱼ)) • A i j` — every coherence is a mode of the flow and its frequency is the eigenvalue gap; `resonanceSpectrum` is the gap set `{λᵢ − λⱼ}`. Stationarity is characterized exactly: fixed at all times **iff supported on the equal-eigenvalue blocks** (`phaseFlow_stationary_iff`). **The narrowing, named:** the commissioned shorthand "stationary iff diagonal" is **false** under degeneracy — a coherence inside an equal-eigenvalue block has gap zero and is stationary without being diagonal; the proved theorem is the block

form, and diagonality is recovered exactly when the generator is nondegenerate (`phaseFlow_stationary_iff_diagonal`; the key step evaluates the phase at t = π/gap, where it is exactly −1). **(III) The thermal instance stands whole:** the corpus's modular flow is the phase flow of K = −ln ρ (`diagFlow_eq_phaseFlow`) — **the resonance spectrum of a state ρ is the gap set of −ln ρ** — and for Gibbs weights the (i, j) coherence rotates at exactly `β·(λᵢ² − λⱼ²)` (`gibbs_resonance_phase`): §19's conversion theorem restated as a frequency law. **Remainder, named:** the proved core is the diagonal generator; reduction of a general Hermitian generator to its eigenbasis form is not formalized (§14, item 18). No continuum field is constructed, and the module is not a claim that these finite operators are electromagnetism.

25. Round 2, closed — the OS re-pin, the Dixmier intake, the BRST intake

The theories intake's Round 2 (queued in §12) ran on 2026-10-18 and is closed. Verdicts are stated per item at the tiers their dossiers rule.

- **The Osterwalder–Schrader re-pin: STANDS at the priced scope.** `OSRepin.lean` (15 declarations, 12 definitions/structures; build green, 3,783 jobs; 12 headlines on exactly the standard three; its three correction theorems depend on **no axioms at all**). Three corrections to the 2026 pin are proved, not asserted. **(1) The premise is stale:** the old header's "Mathlib has no tempered distributions" is false for the pinned Mathlib — $\mathscr{S}'$ ships with Fourier theory — and the opaque `SchwingerMarker` is discharged: Schwinger functions are genuine tempered distributions $\mathscr{S}'$`(Fin n × Fin 4 → ℝ, ℂ)`. **(2) The old reconstruction statement is vacuous:** over bundles whose every content field is an unconstrained `Prop`, the all-`True` instantiation proves it — a theorem with zero analytic content (`oldPin_osReconstruction_trivial`). **(3) The old existence statements are vacuous** the same way (`InteractingOSMeasure`, and its Wightman sibling, §27). The old pin's "open problems" did not state the open problems: they were **unprovable-to-fail**. The re-pin separates the two issues the old shape conflated — the axioms are data plus real quantified conditions (E0 is the $\mathscr{S}'$ type, not a field that could be `False`; E1 Euclidean covariance; E2 reflection positivity; E3 symmetry; E4 clustering), and nontriviality is an explicit clause (`NonGaussian`, the Wick-defect form) — so the re-pinned existence statement `∃ B : OSBundle, NonGaussian B.S` is a real, failable Prop. **It is unproved, and is not claimed.** Clause table, exact: E0 **PINNED**; E1 **PINNED**, with its lattice shadow (discrete motion group) standing; E2 **PARTIAL — the standing shape**: the lattice form stands at every β, positivity at β = 0 stands, and positivity at general β is the conditional Gram criterion whose hypothesis is **open — 1b-B's Gram clause, XL / Mathlib-blocked** (PARTIAL-δ not claimed); E3 **PINNED**, its lattice shadow standing by construction; E4 **PINNED**, β = 0 factorization standing, general-β decay **open = Stage 3/3c**; the lattice shadows of the reconstruction outputs all stand (OS space, vacuum norm 1, transfer contraction, vacuum fixed, T_0 = vacuum projection). Reconstruction itself is pinned as a **schema** with two named analytic holes — Bargmann–Hall–Wightman analytic continuation and edge-of-the-

wedge — both **absent** from the pinned Mathlib (grep-verified) and priced research-scale (§14, item 17). One nuance the re-pin makes visible, recorded not as a defect: E0–E4 plus normalization do **not** exclude the trivial theory — in OS theory that is correct, and interacting content is exactly the extra clause.

- **The Dixmier intake: STANDS at the priced scope — with its prediction corrected.** `DixmierTrace.lean` (419 lines; 24 declarations, 6 definitions; build green, 3,228 jobs; ten headline axiom checks exactly the standard three). The audit is grep-backed: `dixmier`, Banach-limit, and singular-trace terms return **0 hits** in the pinned Mathlib, and `Ultrafilter.lim` does **not** instantiate a Banach limit on ℝ-sequences (compact Hausdorff targets only) — so the generalized limit is pinned as **data** (`GenLimit`: additive, homogeneous, positive, extending ordinary limits; dilation invariance deliberately excluded and named — no statement here needs it), and its existence is **OPEN** (literature, via Hahn–Banach; T4). Proved (T1): the log-mean algebra; vanishing criteria (bounded partial sums, summable, or eventually zero profiles all have trace zero); the **measurability criterion** — if the log-means converge, the Dixmier trace is the limit under every generalized limit (ω-independence); and the **harmonic evaluation**: the profile `n ↦ c/(n+1)` has Dixmier trace exactly **c** under every ω (via Mathlib's Euler–Mascheroni file). **The finite evaluation, and correction C1:** the Dixmier trace of every finite spectrum is **identically zero**, proved — a Dixmier trace is a singular trace and finite-rank operators are its kernel, so at finite level every operator is in the kernel. The gauntlet's predicted "finite, computable shadow" (a residue/trace functional on the finite algebra) was **tested and is wrong**: a finite operator's zeta function is an entire finite Dirichlet sum, with no pole and no residue. The functional that occupies the noncommutative-integral slot at finite level is the **ordinary trace** — which the corpus already owns (`SpectralActionFinite`'s heat coefficients). The FIT verdict stands for the machinery and its (zero) evaluation; as a nonzero functional at finite level it is NO-FIT in principle, not merely unbuilt. **Operator-level Dixmier trace and spectral dimension: ABSENT, NOT CLAIMED** — no corpus operator is `|D|^{-d}` of anything, and no spectral-dimension statement is made for any object.
- **The BRST intake: STANDS at the priced scope — FIT as algebra, NO-FIT as quantization.** `BRSTFinite.lean` (414 lines; 24 declarations, 1 structure and 13 definitions; build green, 3,117 jobs; ten headline axiom checks exactly the standard three). A finite Lie algebra is pinned by structure constants (`FiniteLieData`: antisymmetry, alternation carried as **independent data** — antisymmetry alone does not give it in characteristic 2 — and the Jacobi identity in contracted form); on its Chevalley–Eilenberg cochains the differentials d_1, d_2 are defined by their standard coordinate formulas. **The headline is $d_2 \circ d_1 = 0$, and its proof is literally the Jacobi contraction** — nilpotency in this model is not analogous to Jacobi, it is Jacobi, contracted. The assembled charge `brstQ` on $C^0 \times C^1 \times C^2$ satisfies $Q^2 = 0$, with the bookkeeping direction named in the module (the content-bearing nilpotency is `d₂ ∘ d₁ = 0`). For $\mathfrak{su}$**(2)** in its ε-presentation, everything is computed exactly: d_1 is injective, and **$H^0 = °Ç$ $H^1 = 0$, $H^2 = 0$** (Whitehead vanishing, this finite case; d_2 in fact vanishes on every alternating 2-cochain). The abelian contrast is computed too (zero bracket: every

cochain closed, nothing nonzero exact). **NO-FIT as quantization, stated plainly:** no gauge fixing, no ghost fields, no physical state space, no path integral appears or is claimed — the corpus proves exact gauge invariance without gauge fixing (§13A), and nothing here is a quantization of anything. **Anomalies are ABSENT and named:** no anomaly content is computed anywhere in the brick, and $H^2 = 0$ is a Lie-algebra-cohomology fact, **not** anomaly cancellation (which concerns representation traces). Two scoping fences, named: trivial coefficients and degrees 0–2 only ($\mathfrak{su}(2)$'s $H^3 = \mathbb{C}$ in the literature is not computed here); `d₃ ∘ d₂ = 0` is not proved. **Correction C1, against any future "absent" claim:** the pinned Mathlib does contain an abstract low-degree CE nilpotency (`Algebra/Lie/Cochain.lean`'s `d₂₃_comp_d₁₂`) for extension theory — found by this intake's grep audit (`brst` itself: 0 hits) and recorded; the brick does not route through it, and what is new here is the coordinate form, the assembled charge, and the computed example.

26. The unlock chain — three co-import fences, discharged by renames (repairs; counts invariant)

Three repair landings of 2026-10-17/18, reported together because they are one fact: **no pair of corpus modules is fenced from co-import.** Each repair is renames only, on the bosonic side, by the house pattern of the first; each is certified by the smoke module `FockCoImport.lean`, which now imports both complete second-quantization stacks and both packaging modules into a single environment and proves joint examples there (both number operators kill their vacua; the basis actions side by side). Per-file strict counts are invariant under every repair, and the program total does not move under any of them (2,417 under the first; 3,201 under the second; 3,229 under the third) — a repair that changed a count would not be a repair.

- **The `adag` repair (2026-10-17): STANDS.** The fermionic and bosonic Fock parents both defined `adag`, `ann`, `adag_basis`, `ann_basis` in the shared namespace — four colliding names, different types, so the two stacks could not share an environment and a mixed interaction could not even be stated (§18 is its downstream). The bosonic side was renamed (`adagBos`, `annBos`, and their basis theorems), proved renames-only byte-for-byte. The repair's own audit recorded a correction to its scoping in the open: the four-name repair did not clear the stack-level co-import — a second collision pair, `numberOp` / `numberOp_basis`, blocked it, censused and priced as a separate follow-on rather than improvised into this repair.
- **The `numberOp` rename (2026-10-18): STANDS.** Census verified exactly before any edit: the bosonic side's `numberOp` / `numberOp_basis` — 16 lines in 3 files (11 code occurrences) — renamed to `numberOpBos` / `numberOpBos_basis`; fermionic files untouched; the joint stack import then certified green (3,323 jobs at the certificate's build). The standing collision warnings in five module headers became historical at this landing and were corrected as such by the housekeeping bundle (§0 addendum).

- **The packaging rename (2026-10-18): STANDS.** The fence between the two packaging modules was found by the Fock-Lorentz brick (§22) — and was **six names wide, not one**: the compiler aborts at the first duplicate, so five further collisions (`exists_forall_mem_of_monotone`, `linearCombination_eq_sum` — different statements per statistic under the same name — `ofReal_norm_sq`, `summable_mul_of_summable_sq`, `tsum_eq_of_tendsto_sum_exhaustion`) were invisible until a dedicated census. Twenty bosonic-side lines renamed; the joint-import probe went green on the first iteration; no deeper collisions exist. The repair's dossier records a **declined alternative**: deduplicating the four statement-identical generic helpers into the shared base module was rejected, because it would delete four declarations and move the published total for a pure repair — renaming preserves both the count and the record.

27. The Wightman re-pin — the axioms on genuine $\mathscr{S}$; two vacuity corrections (T1)

`WightmanRepin.lean` (14 declarations, 11 definitions/structures; build green, 3,810 jobs; 13 headlines on exactly the standard three; the existence-vacuity correction depends on **no axioms at all**). No existing file was modified. **No interacting Wightman QFT is claimed to exist.**

- **Correction 1 — the premise is stale (again).** `WightmanAxioms.lean`'s header asserts Mathlib has no Schwartz distribution theory; false for the pinned Mathlib, as in §25. The marker is discharged the same way: Wightman functions are $\mathscr{S}'$`(Fin n × Fin 4 → ℝ, ℂ)`. The old file's genuine T1 content (its light-cone geometry) is load-bearing in the new pin. Its dependency ledger is re-audited in the dossier: item 1 ($\mathscr{S}$) **closed**; item 2 (operator-valued distributions as a general theory) still absent; item 3 (Poincaré representation theory) **partially stale** — the one-particle representation, its Γ-port, and strong continuity of both subgroups at both levels now stand (§22), while a joint full-group object and unbounded-generator theory remain absent; item 4 (reconstruction) pinned as a schema (§25); item 5 (an interacting 4D example) open in mathematics, not merely in formalization.
- **Correction 2 — the old existence statement is vacuous.** `InteractingWightmanQFT` / `WightmanExistence` (`∃ _ : WightmanAxioms, True` over unconstrained `Prop` fields) is provable from the all-`True` instantiation — re-cited here from §25's proof. Unprovable-to-fail, as there.
- **Correction 3 — NEW: Rung 4's sharpened bundle is vacuous as an existence form.** `SharpWightmanTarget` replaced the opaque fields by six sharp definitions with exact formulas — real progress as conditions — but bundled them as an existential over the semantic parameters themselves: the field, the joint spectrum, the spacelike-separation relation, the Wightman functions. Proved (`sharpWightmanTarget_trivial`, T1): junk data satisfies all six clauses at once — H = ℂ, the "field" φ(f) = f • id (scalar multiplication, no operator content), spec = {0}, spacelikeSep = ⊥ — and even W5 falls, since for the scalar-multiple field every point is a length-1 monomial. **What is NOT impugned, stated exactly:** the six

sharp definitions as conditions on fixed, externally given data are exact, and **every Rung 5 verdict against them stands untouched** (§5). What is superseded is the ∃-bundling as an existence form.

- **The re-pin.** `WightmanBundle`: data `W : (n : ℕ) → 𝒮'(Fin n × Fin 4 → ℝ, ℂ)` (temperedness is the type, made explicit by glue theorems), vacuum normalization, **W2** Poincaré covariance (full Lorentz ⋉ translations), Wightman positivity (the GNS input), Hermiticity, **W3 in Fourier-support form** — stated through the plain-Bochner `fourierTest` of the test function, whose integrand is proved integrable, as an annihilator condition on the spectral support (every partial momentum sum in the closed forward cone, total momentum zero) — **W4** locality, and clustering. Vacuum cyclicity is deliberately **not** a bundle field: it is reconstruction content, whose distribution-level input is positivity — the same separation §25 made. Nontriviality is separated identically: `InteractingWightmanStatement := ∃ B : WightmanBundle, NonGaussian B.W` — real, failable (a proof that every bundle is Gaussian would refute it), and **UNPROVED**. `SatisfiesWightmanPin` fills the OS reconstruction schema's second hole with a concrete predicate; the first hole — analytic continuation — stays research-scale (§25, §14 item 17).
- **The window table, at the one-mode fermionic model** (that scope is part of each verdict): **W1 PROVED** at the model; in general **OPEN** — and the blocker has moved: the test-function side is no longer it (Correction 1); the field construction is. **W2 PARTIAL** — $\mathbb{Z}_2$ parity at the model; its price is narrowed by §22 (the representation, its port, and strong continuity stand; what remains is the covariance equation for an actual field, the geometric action on spacetime test functions, and any interacting case). **W3 PARTIAL** — frame consistency at the model; the re-pin states the general clause completely in Fourier-support form, while proving things with it needs Stone's theorem and joint spectral calculus for unbounded operators, absent from Mathlib (the corpus has finite-dimensional Stone only). **W4 PROVED** at the model (algebraic content; the model has no spacetime); in general OPEN. **W5 REFUTED as framed, corrected form PROVED** — closed at the model in the literature's meaning (§5). **W6**: the bosonic form at the model is a documented **FALSE POSITIVE** (it certifies the free Majorana field as nontrivial); the corrected Koszul–Wick form is PROVED. **Scope fence:** the bundle pins the **scalar** form of the axioms; a spin refinement and a signed fermionic defect clause are unbuilt and named.

28. Two exhibits of a different kind — the trit experiment (a kill), and the vacuum portrait (samples)

Neither exhibit is a Lean theorem. Both are reported here because the program ran them under the same discipline: pre-registered verdicts, fences printed, negative results at the same speed as positive ones.

- **The trit experiment — DIES-K3 (a complete kill, reported as a result).** A pre-registered software experiment (protocol written before any scored run) testing whether ternary weight representation is a separable energy lever for

inference — the empirical premise of a proposed data-center direction. Word-level language model on WikiText-2 (4,870,912 parameters; fixed accuracy band: perplexity ≤ 784.04). In band: FP16 (712.76), INT8, INT4 (717.58), and quantization-aware-trained ternary (696.13); out of band: INT2 and post-training ternary. On the modeled energy proxy (an op/traffic census against a published 45 nm table — a **model**, labeled as such; the wall-energy proxy was environment-blocked and the verdict rests on the modeled proxy with that fence printed), ternary beat the best binary baseline (INT4, 404.602 µJ/token) by **+58.03%** (169.800 µJ/token) — passing the pre-registered margin test. **The kill is the decomposition:** weight traffic is 96–98% of every total, and a hypothetical binary scheme at exactly ternary's bit-width posts 169.692 µJ/token — the radix-attributable residual is **−0.108 µJ/token ≤ 0**, so the pre-registered K3 fires: **it is the bits, not the base.** At the 2-bit packing real ternary kernels use, the two are indistinguishable (210.53 vs 210.45 µJ/token). What survives, exactly: low precision helps; three-ness does not separably help. **Tier discipline:** no ternary hardware exists and none was measured; no hardware energy saving, no facility percentage, and no water figure is licensed by anything here. The direction the experiment was chartered to test is dead in writing, by the program that proposed it — the difference between a program that formulates and one that advertises.

- **The vacuum portrait — an exhibit of samples, with its fences.** Six panels sampling the corpus's own Stage 0/1a definitions: the U(1) Gibbs measure (circle group, fundamental representation) on a 6^4 periodic torus, by exact sampling at $\beta = 0$ and heatbath above it (fixed seed 20261018; 300 thermalization sweeps, 3,000 configurations per β; batch-means errors). Every pixel is traceable to a Lean definition — configuration, plaquette, Wilson action, Gibbs weight are the corpus's objects, not an independent model's. **Anchors against theorems:** $\langle \mathrm{Re}\, U_p \rangle$ at $\beta = 0$ measures $+0.00013 \pm 0.00012$ against the proved value exactly 0; the $\beta = 0$ Wilson loops $R = 1, 2, 3$ all measure within ± 0.0003 of the proved 0; and at $\beta = 0.25$ the sampler measures 0.12404 against §21's proved first-order value 0.125. The curve: 0.24566 at $\beta = 0.5$; 0.60521 at $\beta = 1.0$; 0.86502 at $\beta = 2.0$; 0.91268 at $\beta = 3.0$. The difference portrait ($\beta = 1.0$ minus the $\beta = 3.0$ vacuum reference) is near-flat — range $[+0.2833, +0.3231]$ about 0.3075 — which is itself the finding, and is forced by translation invariance: with nothing inserted, the thermal excitation content of this toy is delocalized. The flux-tube panel enters external charges the only way a pure-gauge theory honestly can, as a static Wilson loop: conditioned on the 3×3 loop ($\langle \mathrm{Re}\, W(3\times3) \rangle = +0.5173 \pm 0.0110$ at $\beta = 2.0$), the action-density dip inside the loop peaks at $|\Delta| = 0.0176$, ≈ 5× the noise floor — with the **abelian Stokes caveat printed on the exhibit**: in an abelian theory the loop is exactly the product of its interior plaquettes, so the dip is that correlation made visible, not a nonabelian flux tube's energy profile. **One blemish, disclosed on the exhibit:** $\beta = 1.0$ sits in the finite-volume crossover — an independent cold-start run gives 0.5946 ± 0.0034 against the production 0.6052 ± 0.0046; read that point's true uncertainty as ≈ 0.01. **Fences, printed on the exhibit and repeated here: U(1) only — not SU(3), not QCD; an L = 6 toy torus at fixed lattice spacing — no continuum limit is taken or**

implied; every portrait is a 2D slice of a 4D object; samples are not the theory (a configuration is one draw; the theory is the measure); pure gauge — the "electron cloud" enters only as static external charges, and a dynamical electron-cloud exhibit is a separate, unbuilt item; β = 3.0 is a vacuum reference, not T = 0; and no mass-gap claim of any kind. The exhibit is evidence of nothing beyond its anchors: it is a portrait of definitions the corpus proves things about, drawn so the measure can be seen — and, per §17, an independent re-run of its pipeline reproduced every number and panel byte-identically.

29. The 4D Dynamics in four colors — a visual representation (exhibit)

This section is an exhibit, like §28: it proves nothing and adds no theorem. It sets four equations of the corpus — each machine-checked in the module named beneath it — in four colors, as a visual representation of the program's dynamic ontology: heat in a bounded box; a thermal state resting on its energies; charge separated across sectors; the flux that answers when sectors couple; and the state itself, whose snapshot is the algebra and whose flow is the film behind it. The equations are T1 exactly as stated in their modules. The colors are a reading aid under a palette declared below; the ontology the palette encodes is the author's interpretive layer (T5 in the program's tiers) and asserts nothing about the mathematics, which stands in black and white in the repository. Each strand is named in its caption, so nothing in this section depends on color alone surviving print.

The palette (declared convention of this exhibit). Gold — heat and diffusion. Blue — charge and its separation. Red — the modular flux. Green — the state and its flow. The assignment extends the author's standing color convention for the tripotent components (X red, Y blue, Z green) with a fourth color for heat; it is a convention of representation, not a claim.

- **$E[r^2] = x^2 + y^2 + t$** — gold: heat, spreading. The expected squared displacement of the two-dimensional walk after t steps from (x, y) is exactly $x^2 + y^2 + t$: time enters the distance law in quadrature with the two spatial directions, and diffusion — the heat channel — spreads at this, the slowest of rates. `ThetLogos.DiffusionDefect` (12 theorems, zero sorry).

- **$S(\beta) = q \cdot (\exp(-\beta E_0) - \exp(-\beta E_1)) / (\exp(-\beta E_0) + \exp(-\beta E_1)) = -q \cdot \tanh(\beta(E_0 - E_1)/2)$** — blue: charge, separated by a thermal state. For a Gibbs state over two sectors with energies E_0, E_1 and an input charge q carried as +q on sector 0 and −q on sector 1, the separation is exactly this function of the inverse temperature β and the gap: zero if and only if the energies are equal, strictly growing in β for a nonzero gap, bounded by |q| at every finite β, and tending to ±q in the cold limit. The charge and the energies are inputs; no charge value is derived. `ThetLogos.TeslaVortexSeparation` (18 theorems, zero sorry; landed 2026-10-22, after this manuscript's §0 inventory — its dossier is `whitepaper/tesla-vortex-modular-dynamics-rung1-2026-10-22.md`).

- **φ = i[K, D] = 0 ⟺ D is block-diagonal across ρ's distinct-weight sectors** — red: the flux. The modular flux of a state vanishes exactly when the Dirac operator does not couple the state's weight sectors — for arbitrary D and any real weights, with the general Hermitian form beside it. Couple the sectors of a separated state, and the flux is nonzero: the vortex turns on. `ThetLogos.FluxClassification` (15 theorems, zero sorry; landed 2026-10-22; dossier `whitepaper/flux-classification-2026-10-22.md`).

- **K = β•diag(E) + logZ•1, and the modular flow of a Gibbs state is Heisenberg evolution at rescaled time β·s** — green: the state and its flow. For a Gibbs state the modular Hamiltonian is the energy diagonal, rescaled: the snapshot's generator is written on its face, and its flow is ordinary evolution read at thermal time — in both pictures, for a general Hermitian Hamiltonian. `ThetLogos.GibbsSpan` (21 theorems, zero sorry; landed 2026-10-22; dossier `whitepaper/gibbs-span-2026-10-22.md`).

Fences. The four equations are the whole content of this exhibit. Nothing here models light as a field, matter formation, or the derivation of any physical constant or charge; the map from these equations to the author's 4D Dynamic Ontology is printed as a separate memo (`4d-dynamic-ontology-combined-map-2026-10-22`, outside this paper) with every correspondence marked anchored, rhyme, or empty, and the standing kills restated. A reader who grants the ontology nothing loses nothing here: four proved equations, in colors.

Acknowledgments

This program formalizes. It did not originate the ideas, the problems, or most of the mathematics it verifies. The staircase this corpus stands on, named at the author's direction ("make sure the true credit goes to them"; the repository's `ACKNOWLEDGMENTS.md` carries the same text):

- **The problem.** Chen Ning Yang and Robert Mills — the gauge theory whose rigorous construction is the summit this program climbs toward. Arthur Jaffe and Edward Witten — the official Clay statement of the Yang–Mills existence and mass gap problem. The Clay Mathematics Institute — for posing it as a Millennium Problem.
- **The geometry.** Paul Dirac — the Dirac operator at the heart of every spectral triple in this corpus. Alain Connes — noncommutative geometry and the spectral triple; the distance formula, the spectral action, and the framework this corpus formalizes. Ali Chamseddine — co-architect of the spectral action and of the almost-commutative Standard Model geometry.
- **The modular theory.** Minoru Tomita and Masamichi Takesaki — the modular theory that turns a state into a flow. Alain Connes and Carlo Rovelli — the thermal time hypothesis.
- **The axioms and the fields.** Arthur Wightman — the axioms a rigorous quantum field theory must satisfy. Konrad Osterwalder and Robert Schrader — the Euclidean axioms and reconstruction. Werner Heisenberg — the operator

language of quantum mechanics. Erwin Schrödinger — the equation whose dynamics this corpus formalizes. James Clerk Maxwell — the first gauge field theory. Albert Einstein — the field equations whose Hilbert-type terms the spectral action recovers at finite level. Isaac Newton and Gottfried Wilhelm Leibniz — the calculus and the limit: the continuum this program works to reach honestly.

- **The lattice.** Kenneth Wilson — lattice gauge theory and the renormalization group picture; the route §13A takes. Shmuel Elitzur — the theorem that gauge-variant quantities cannot acquire expectation values; proved here as a result, not assumed as a slogan. Miroslav Fiedler — the variational eigenvalue bound used in this corpus's graph-Laplacian exhibit. Alfréd Haar — the measure every lattice expectation here is integrated against.
- **The analysis.** Laurent Schwartz — distributions and the function spaces of this corpus's analytic rungs. Marshall Stone and John von Neumann — the spectral and unitary machinery underneath the dynamics. Leonhard Euler — the action-and-extremum tradition the whole Lagrangian world descends from.
- **The staircase in modern form.** The Lean community and the Mathlib contributors — thousands of people whose verified library this corpus imports directly; our theorems stand on their theorems, mechanically and literally. The constructive field theory tradition — Glimm, Jaffe, Balaban, and many others — whose decades of work mark the frontier this program measures itself against.

Formalization, verification, and the finite-level results are ours. The concepts are theirs. If this corpus is ever cited, cite the originators first.

Appendix A. Background algebra (fenced — not part of the ladder)

This appendix reports five T1 modules that stand alone. They are **not** rungs of the ladder, **not** physics, and **not** claimed to connect to the corpus's operator-algebra structures; they are recorded here so the inventory is complete and the fence is explicit.

- **Ternary substrate (`TritGrading.lean`, 350 lines; T1).** A three-element type `Trit` (`neg | zero | pos`) with its full abelian- group laws proved by case analysis and a bundled additive equivalence `Trit.addEquivZMod3 : Trit ≃+ ZMod 3`; a submodule-valued grading structure `TritGrading` whose only axiom is multiplication closure, with commutator closure (`TritGrading.commutator_mem`) and adjoint- action closure (`TritGrading.ad_mem`) proved, plus a block-offset instance; honestly named commutator facts (`commutator`, `ad` — not "curvature", not "covariant derivative"); and the squared-commutator trace bound `trace_commutator_sq_nonpos` for Hermitian matrices (`trace [A,B]²` is not a Chern character, and is not named as one). Zero sorrys; headline axiom footprint exactly `[propext, Classical.choice, Quot.sound]`. **Connection audit (honest): none.** The corpus's "ternary" is the TRO product `⟨a, b, c⟩ = a · star b · c` — a 3-ary operation, a different notion of "three" from a cyclic grading; nothing in the corpus grades the TRO or its tripotents by $\mathbb{Z}_3$, and the corpus's operator-algebra gradings are $\mathbb{Z}_2$ (the spectral

triple grading `γ_F`). The substrate stands alone as background algebra: hygiene and readiness, no forced bridges.

- **Sexagesimal clock (`SexagesimalClock.lean`, 433 lines; 31 theorem/lemma declarations, 12 definitions, 1 structure; T1).** The $\mathbb{Z}_{60}$ sibling of the trit module, built to the same standard: unique two-place sexagesimal representation for every n < 3600 (and the three-place version for n < 216000), both with uniqueness proved; the divisibility core (every k in {1,2,3,4,5,6} divides 60, and lcm(1,...,6) = 60 — the last depending on no axioms at all); the canonical sub-clock reduction maps out of $\mathbb{Z}_{60}$, the mod-3 one named `toTritClock`; the Chinese remainder decomposition `crtEquiv : ZMod 60 ≃+* ZMod 3 × ZMod 4 × ZMod 5` with the projection compatibility `crtEquiv_fst` proved (the first component is the trit reduction); a `ClockGrading` structure with commutator and adjoint-action closure; and the headline `ClockGrading.reduceToTrit` — every $\mathbb{Z}_{60}$ clock grading pushes forward along the mod-3 reduction to a genuine `TritGrading`. That is grading bookkeeping: the trit clock sits inside the sexagesimal clock as gradings. Not a physical identification. Targeted build green (3,118 jobs); zero `sorry`, `admit`, or `native_decide`; eight headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`. One vocabulary fence, stated in the file header: the "modular" of this program is Tomita modular theory; this module is modular arithmetic, and the rhyme earns nothing.
- **Flux-twist toy (`FluxTwist.lean`, 582 lines; 38 theorem/lemma declarations, 10 definitions; T1).** A finite toy on the same shelf: the twisted shift on the ring of `Fin (n+1)` sites, the boundary bond carrying the flux phase. Proved: unitarity (`twistedShift_unitary`, both products with the conjugate transpose); flux periodicity (`twistedShift_add_two_pi`: S(θ + 2π) = S(θ)); gauge redistribution (`twistedShift_mul_gaugeMatrix` — the boundary twist is the uniform twist, conjugated by the diagonal gauge matrix whose site-k phase is θ·k/(n+1)); and the ring hearing the flux — the eigenphase (θ + 2πm)/(n+1), proved for the uniform twist on plain Fourier vectors and for the boundary twist on flux-carrying Fourier vectors (plain ones fail the wrap bond by exactly the flux factor; recorded in the file). The stretch target landed except its priced item: the hopping Hamiltonian is Hermitian (`hopping_isHermitian`) with eigenvalues −cos((θ + 2πm)/(n+1)); the Gibbs-state modular-flow generation statement is **priced, not formalized** (it needs matrix exponentials and the finite-dimensional modular theory of Gibbs states), and is stated as priced in the file header. The vocabulary fence is the module's point: "magnetism" appears only as flux twist — the Aharonov–Bohm effect in miniature. It is **not** a bridge to real magnetism or Maxwell's equations, and **not** a Yang–Mills rung. Targeted build green (3,120 jobs); zero `sorry`, `admit`, or `native_decide` in code; six headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`.
- **Weld duality (`WeldDuality.lean`, 893 lines; 73 theorem/lemma declarations, 18 definitions; T1).** Finite linear algebra: the finite shadow of the Morley idea of §13, built in both of its configurations as two faces of one module. The common core: for an involution η (η² = 1), the spectral projections `weldPlus` / `weldMinus` = (1±η)/2 are complementary orthogonal idempotents, η = $P_+ - P_-$, the trace formula `tr P₊ = (n + tr η)/2` holds, and the signature count is rigid under

conjugation. Face A is the grading split with a corrected involution: `corrInv` in parametrized form (an involution when $\eta^2 = A^2 = 1$, $\eta A + A\eta = 0$, $c^2 = 2$), and on the $\mathbb{C}^4$ arena ($\eta = \mathrm{diag}(1,1,1,-1)$) a corrected involution carrying a Hadamard block on the mixed directions (`etaCorr_sq`). Face B is the Dirac sign with the gap split: for $D = \mathrm{diag}(2,1,-1,-3)$, the sign involution η_D is Hermitian with $\eta_D^2 = 1$, its plus-projection is the positive spectral subspace by computation (`weldPlus_etaDirac`), and $H = D^2$ is split at $\Delta^2 = 2$ into complementary orthogonal projections with `gap_dichotomy` (every eigenvalue is 1 or ≥ 2). The linkers are proved, and they are what makes the two faces one module: Face B is an instance of Face A (`signature_etaDirac`), so the signature count follows the **input** in both faces — the §13 kill (`signature_not_determined`) stands untouched, and nothing here derives (3,1); and the curved correction leaves the plus-projection's trace invariant while the projections genuinely differ (`trace_weldPlus_etaCorr`, `weldPlus_etaCorr_ne`) — the stability is not vacuity. The resting-place theorems the module was designed around are proved: the fixed set of an involution is exactly the range of its plus-projection (`fixed_iff_range_weldPlus`), and for two commuting involutions the joint fixed set is the range of the product projection (`joint_fixed_iff_range`). **Priced, not attempted:** the general correction-stability theorem (the plus-projection's count constant along a general correction, for arbitrary n) — only the worked-arena instance above is proved. The fence, stated in the file header: "2D holography" does not survive into this module — only the finite shadow does, dimension counted by traces of spectral projections. **Not** a Yang–Mills rung. `EdgeSignature.lean` is not imported (its §1 pair algebra is restated locally; disclosed in the file header). Targeted build green (3,120 jobs); zero `sorry`, `admit`, or `native_decide` in code; seven headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`.

- **Graph-Laplacian bridge exhibit (`BridgeLaplacian.lean`, 466 lines; 29 theorem/lemma declarations, 7 definitions; T1).** (Added in Version 3.) On a finite weighted graph (vertex type V, symmetric weights w as bare data — symmetry and nonnegativity are theorem hypotheses, never baked into the definitions): the bridge operator `dropVec F i j = F j - F i` between node potentials and edge drops, with its linearity, antisymmetry, and the telescoping path law whose closed-path corollary is the discrete Kirchhoff voltage law; Ohm's law as the middle factor `ohm w X i j = w i j * X i j`; and the **factorization** `laplacianOp_eq_diverge_ohm_drop` — $L = d^T \sigma d$ — a genuine theorem, not a definition (the divergence gathers transposed entries, so the two sides agree only after the symmetry of w is spent). With it: the current law ($\sum_i (LF)_i = 0$), the dissipation identity ($2\cdot\text{energy} = \sum_{ij} w\,i\,j\,(F_i - F_j)^2$) and energy nonnegativity, self-adjointness of the form, the kernel characterization (zero energy ⟺ F constant across every conducting edge; at all-positive weights, F constant), and the **Fiedler value** as a variational infimum with its bound proved whole (`fiedler_le_energy`, `fiedler_nonneg`) — the pre-authorized fallback of pricing the bound was not needed. The three readings of the exhibit (resistor network / graph heat / DeGroot consensus) are recorded as an equation identity — one operator equation read three ways — claiming nothing about real circuits, real heat, or real electorates. **Priced, not attempted** (at Version 3; all three DISCHARGED in

Version 4 by *`BridgeCompletion.lean`*, §23: attainment of the variational infimum, the minimizer's eigenvector property with the λ_2-identification, exponential decay at the Fiedler rate, and the graph-Faraday package — the last fenced there as bookkeeping, not Maxwell theory). Relation to Mathlib, stated in the dossier: Mathlib's `SimpleGraph.lapMatrix` is the unweighted 0/1 Laplacian; this exhibit differs in kind (arbitrary symmetric real weights, function-level operators, the explicit $d/\sigma/d^T$ factorization), and no novelty beyond that packaging is claimed. Not a Yang–Mills rung, and not connected to §13A's lattice: graph and lattice objects differ in kind, and no bridge between the exhibit and the lattice stages is claimed. Targeted build green (3,117 jobs); zero `sorry`/`admit` in code; eight headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`.

Notation. φ denotes the modular flux i[K, D] (Hermitian, `modularFlux_selfAdjoint`). Φ is reserved for the ontological glyph and is not used in this paper. Tet (the narrator of the companion books) and thet (the law) are distinct; neither is a mathematical object of this paper.